Multiscale Spectral–Spatial Coordinated Awareness Fusion Network for Hyperspectral Image Classification

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***Abstract*—Accurate hyperspectral image (HSI) classification remains challenging, as discriminative performance critically depends on the effective coordination of high-dimensional spectral information and complex spatial structures. Recent deep learning (DL) methods, particularly attention mechanisms, have improved HSI classification. However, existing models still face spectral over-smoothing, spatial ambiguity, and poor spectral–spatial coordination. To address these limitations, a multiscale spectral–spatial coordinated awareness fusion network (MS^{2}CAFNet) is proposed for HSI classification. The proposed framework adopts a dual-branch architecture to explicitly decouple spectral and spatial feature learning, thereby alleviating cross-domain interference. In the spectral branch, a difference-gated multiscale spectral attention (DGMSA) module is designed to jointly model spectral similarity and difference cues under multiple receptive fields, enabling an adaptive balance between redundancy suppression and fine-grained discrimination. In the spatial branch, a multihead coordinate-aware spatial attention (MHCASA) module is introduced, which integrates explicit positional encoding with long-range contextual modeling to enhance robustness in spatially heterogeneous scenes. Furthermore, a bidirectional cross-guided fusion (BCGF) strategy is developed to align and coordinate spectral and spatial representations through correlation-based interaction, ensuring semantic correspondence across branches. Experimental results on four publicly available HSI datasets demonstrate that the network outperforms state-of-the-art methods.**

***Index Terms*—Attention mechanism, dual-branch network, hyperspectral image (HSI) classification, multiscale modeling, spectral–spatial feature learning (SSFL).**

I. Introduction

With the rapid development of satellite- and airborne-based remote sensing technologies, hyperspectral images (HSIs) have attracted increasing attention due to their ability to record hundreds of contiguous spectral bands ranging from the visible to short-wave infrared regions at each pixel location [1], [2]. Unlike conventional remote sensing images, HSIs contain both high-dimensional spectral data and fine spatial structures. They can generate highly discriminative spectral signatures. Owing to these advantages, HSIs have been extensively applied in a variety of domains, including precision agriculture [3], [4], mineral exploration, environmental monitoring [5], [6], and urban planning [7], [8]. In these applications, accurately distinguishing and classifying different land-cover targets serves as a fundamental prerequisite for higher level analysis tasks. Consequently, HSI classification has become a critical research topic in the field of remote sensing [9]. Multimodal remote sensing data fusion has also been investigated to enhance HSI quality by integrating complementary information from multiple sensors [10].

Beyond HSI classification, recent studies have explored related tasks such as hyperspectral video tracking and anomaly detection, further highlighting the importance of effective spectral–spatial representation learning. In hyperspectral tracking, models such as SASU-Net [11], and SiamSTU [12] integrate spectral adaptive aggregation and spatial–spectral interaction mechanisms to capture temporal dynamics and improve the robustness of target localization. In addition, cross-modal spectral complementarity frameworks [13] utilize memory and multimodal cues to enhance feature representations under complex scene changes. In hyperspectral anomaly detection, recent studies [14], [15] have focused on identifying subtle spectral deviations from the background by utilizing local contrast, residual learning, and background estimation strategies. These methods emphasize the importance of preserving fine-grained spectral differences, which is also crucial for improving classification performance in scenes with high interclass similarity. Although these studies are designed for different tasks, they consistently demonstrate that effective modeling of spectral variations, spatial structure, and cross-domain interactions is fundamental to hyperspectral analysis, which further drives the design of advanced spectral–spatial learning frameworks for classification.

In the early stages of HSI classification research, conventional machine learning (ML) methods were widely adopted, such as k-nearest neighbors (KNN) [16], support vector machines (SVMs) [17], Markov random fields (MRFs) [18], and Gabor filter [19]. However, these approaches typically rely on handcrafted feature extraction and are therefore limited

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in their ability to fully exploit the complex spatial structures and high-dimensional spectral characteristics inherent in HSIs. As a result, their performance degrades noticeably in complex scenes with high interclass similarity, which fundamentally restricts further improvements in classification accuracy.

In recent years, deep learning (DL) has achieved remarkable progress in HSI classification owing to its powerful feature representation and nonlinear modeling capabilities [20], [21]. Li et al. [22] proposed a model-driven DL framework that integrates physical degradation models into deep networks for HSI reconstruction. However, many DL methods rely on large training datasets, which limits their application in real-world scenarios. To alleviate this problem, Li et al. [23] explored deep prior-based methods for HSI reconstruction. Among various architectures, convolutional neural networks (CNNs) have emerged as a mainstream framework for HSI classification due to their structural regularity and parameter-sharing properties [24], [25]. Multiscale convolutional structures, by incorporating kernels with different receptive fields, can enhance the model's ability to represent features at multiple levels, capturing both local texture details and global structural patterns. For instance, Gong et al. [26] proposed a multiscale CNN (MSCNN), which significantly improves the expressive power of HSI feature representations. Pan et al. [27] developed a multiscale hierarchical inverted-pyramid module, which strengthens the complementarity between local and global features by integrating features at various scales across adjacent branches.

However, although CNNs can hierarchically extract local spectral–spatial features, their limited receptive fields restrict the modeling of long-range contextual dependencies and may result in the loss of critical fine-grained information [28]. Recently, attention mechanisms have been widely adopted in HSI classification due to their ability to model long-range dependencies effectively [29], [30]. By highlighting discriminative features and suppressing redundant information, attention mechanisms enable adaptive modeling of crucial spectral bands or spatial regions [31]. For example, Mou and Zhu [32] developed a spectral attention module-based convolutional network, which adaptively learns the weights of individual spectral bands, significantly enhancing spectral feature discriminability. Ding et al. [33] designed a local attention network that embeds attention into high-level features to strengthen the semantic information of low-level features, mitigating discrepancies between physical information and spatial distribution. Hang et al. [34] proposed an attention-aided CNN model that leverages the discriminative priors of both spectral bands and spatial locations, adaptively assigning higher weights to highly informative spectral responses and spatial neighborhood features, thereby effectively suppressing redundant information. However, many attention-based methods mainly rely on similarity-driven feature aggregation, which may lead to the spectral over-smoothing problem. In HSI feature learning, spectral over-smoothing refers to the phenomenon where repeated similarity-driven feature aggregation causes the spectral responses of different materials to become excessively homogeneous in the learned feature space [35]. As a result, fine-grained discriminative spectral variations, particularly at class boundaries or subtle material differences, are progressively weakened.

Despite these advances, attention-based models still face critical challenges as follows.

1) Existing spectral attention mechanisms generally suffer from a common flaw: their modeling focus is overly biased toward mining global similarities between spectral bands to filter redundant information, while neglecting subtle differences in discriminative value between bands. This phenomenon reduces the discriminative power of spectral features and may negatively impact classification performance.
2) Existing spatial attention mechanisms often focus on capturing global spatial relationships during the modeling process, while weakening the representation of local neighboring pixels, and inherently lack explicit spatial coordinate perception capabilities. This deficiency makes it difficult for the model to distinguish similar texture patterns in different spatial locations, ultimately limiting the discriminative representation efficiency of spatial features.

The above limitations indicate that, despite their effectiveness, existing attention mechanisms mainly operate by refining feature responses within a shared representation space. However, HSIs exhibit fundamentally different statistical properties and discriminative information along the spectral and spatial dimensions. Strong coupling of spectral and spatial features in a single space causes mutual interference. This limits the performance of attention-based refinement. To improve HSI classification, we need better attention modules and a better architecture for spectral–spatial feature extraction, organization, and interaction. Motivated by this observation, several studies have attempted to alleviate spectral–spatial interference by adopting dual-branch architectures, where spectral and spatial features are processed separately before fusion.

In recent years, numerous studies have explored dual-branch architectures. For instance, Mei et al. [36] proposed a spectral–spatial attention network to address the sequential dependency of spectral information and the local structural characteristics of spatial information in HSIs. Their network adopts a parallel dual-branch architecture, where the spectral branch incorporates recurrent neural networks (RNNs) to capture long-range interband dependencies and deeply mine spectral sequence features, while the spatial branch leverages the local receptive field properties of CNNs to accurately extract spatial structures and texture features of ground objects. Cui et al. [37] proposed a dual-branch local context feature extraction network, in which the spectral branch embeds a multiscale dense convolution module to facilitate the reuse and fusion of shallow and deep spectral features through dense connections. Meanwhile, the spatial branch employs self-calibrated convolution to establish long-distance pixel dependencies, enabling precise modeling of spatial topology and neighborhood contextual information. Li et al. [38] proposed a dual-branch network with dual attention mechanisms, where channel attention blocks and spatial attention blocks are, respectively, applied to the spectral and spatial branches to optimize feature extraction. While these methods explicitly

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decouple spectral and spatial modeling, many existing fusion approaches rely on simple operations such as feature splicing or summation. These direct fusion strategies treat different feature branches independently and do not explicitly model the interactions between heterogeneous features. Therefore, the complementary relationships between multiscale features are not fully utilized, which may limit the model's ability to highlight discriminative spectral features. Several studies have explored interaction-based fusion strategies to improve feature fusion in remote sensing tasks [39]. Cross-modal learning strategies have also been used to mine complementary information between heterogeneous remote sensing data [40]. Jing et al. [41] presented a dual-branch attention convolutional spectral–spatial feature extraction network, which combines attention convolution and spectral–spatial attention mechanisms to dynamically extract multiscale features while alleviating the limitations of long-range spatial dependency modeling, followed by feature fusion across branches. Wu et al. [42] proposed a dual-branch spectral–spatial large-kernel attention network, which uses a feature fusion strategy to deeply fuse attention-optimized spectral discrimination features and spatial topological features to improve classification performance.

Although these dual-branch network architectures have achieved promising classification results, they still face critical challenges as follows.

1) Spectral and spatial features exhibit inherent semantic correspondence. Without explicit cross-branch guidance and alignment mechanisms, naive feature fusion may fail to realize complementary enhancement and may even introduce redundant interference, thereby degrading the discriminative capability of the fused representations.
2) More importantly, existing designs generally lack a unified coordination mechanism, making it difficult to achieve effective alignment and collaborative representation of spectral and spatial features across multiple scales and semantic levels. This limitation becomes particularly pronounced in complex land-cover scenarios, directly constraining the performance ceiling of dual-branch architectures in HSI classification tasks.

To address the aforementioned challenges, we propose a multiscale spectral–spatial coordinated awareness fusion network (MS^{2}CAFNet). Instead of synchronously modeling spectral and spatial information within a unified feature space, MS^{2}CAFNet adopts a dual-branch architecture to explicitly decouple spectral and spatial representations, thereby reducing cross-domain interference and enabling targeted feature modeling. In the spectral branch, multiscale modeling is employed to capture discriminative information under different receptive fields, while both spectral similarity and difference cues are explicitly characterized and adaptively balanced to avoid over-smoothing and preserve fine-grained spectral discrimination. In parallel, the spatial branch focuses on modeling complex spatial heterogeneity by jointly considering global contextual dependencies, local neighborhood structures, and explicit positional information through a coordinate-aware attention mechanism. To further ensure effective interaction between heterogeneous representations, a bidirectional cross-guided fusion (BCGF) strategy is introduced to align spectral and spatial features via correlation-based interaction, enabling coordinated perception and complementary enhancement. Through the above design, MS^{2}CAFNet achieves separate yet coordinated modeling of spectral and spatial features and realizes deep interaction and alignment across multiple scales and semantic levels, leading to improved discriminative representation and robust classification performance in complex scenes. The main contributions of this work are summarized as follows.

1) A difference-gated multiscale spectral attention (DGMSA) module is designed, which explicitly explores spectral similarity and difference cues under multiple receptive fields. By adaptively balancing these complementary cues via a gating mechanism, the proposed module mitigates over-smoothing effects caused by similarity-dominant attention and enhances fine-grained spectral discriminability.
2) A multihead coordinate-aware spatial attention (MHCASA) module is proposed, which jointly encodes absolute positional information and long-range spatial dependencies. The proposed module integrates coordinate-aware convolution (CoordConv) with multihead attention and local convolutional pathways. This design captures both global contextual structures and local geometric details, improving robustness in spatially heterogeneous scenes.
3) A BCGF module is developed to model the interdependency between spectral and spatial representations. Through constructing a bidirectional correlation matrix, the proposed module enables alignment-based interaction between the two branches, ensuring semantic correspondence while enhancing the discriminative capacity of the fused features.
4) The MS^{2}CAFNet is proposed. The network adopts a dual-branch architecture to decouple spectral and spatial representations and enables coordinated modeling through an alignment interaction mechanism, thereby alleviating cross-domain interference and enhancing discriminative feature learning.

The rest of this article is structured as follows. Section II reviews the related works of the proposed network. Section III introduces the proposed MS^{2}CAFNet method. Section IV describes the datasets, experimental parameter design, and classification accuracy comparisons. Finally, Section V concludes this article.

II. Related Work

Motivated by the aforementioned challenges in spectral discrimination, spatial dependency modeling, and spectral–spatial coordination, this section reviews representative studies from three aspects: spectral modeling, spatial modeling, and dual-branch spectral–spatial fusion.

A. Spectral Modeling Methods

Accurate modeling of spectral information is fundamental to HSI classification, because the spectral dimension provides rich material-dependent signatures that are critical for

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discriminating land-cover categories. Early DL-based studies focused on exploiting spectral correlations by treating each pixel as a spectral sequence and learning band-wise dependencies through hierarchical representations. For example, He et al. [43] proposed a multiscale 3-D deep CNN to hierarchically capture spectral characteristics at different levels, demonstrating the effectiveness of deep spectral feature learning for HSI classification. Zhong et al. [44] proposed a spectral–spatial residual network that further introduced residual learning, enabling deeper architectures while preserving spectral discriminability. Gao et al. [45] used a novel architecture combining alternating small convolutions and feature reuse to address the problem of spectral band correlation. This architecture not only effectively weakens the impact of strong interband correlation on classification, but also fully exploits the spectral information of HSI. Azar et al. [46] proposed a sparse modeling framework for HSI classification, which employs a sparse structure to extract the most discriminative spectral blocks. Xu et al. [47] proposed a local–global spectral feature extraction and optimization method, which enhances the correlation between spectral features and expands the diversity of feature morphology. Han et al. [48] proposed the subpixel spectral variability network to address the existence of mixed pixels and spectral heterogeneity. This network integrates subpixel and spectral cosine correlation information, thereby improving the network's classification performance. However, these studies indicate that most existing spectral modeling methods rely primarily on similarity-driven aggregation or single-type correlation modeling, which can lead to overly smoothed spectral responses in the learned feature space. Consequently, subtle interclass spectral differences may be suppressed, especially near class boundaries, where discriminative spectral differences are relatively weak.

B. Spatial Modeling Methods

In addition to spectral information, spatial context plays a vital role in HSI classification by providing structural and geometric cues that help resolve spectral ambiguity. Duan et al. [49] proposed a modeling method that integrates dual spatial information. They introduced the structure profile (SP), constructed via adaptive texture smoothing, for remote sensing feature extraction. Combined with an extended random walk to optimize spatial information, they significantly improved classification performance by fusing two-stage classification probabilities through decision fusion. To effectively model spatial context information, Jiang et al. [50] proposed a convolutional spatial propagation network, which captures the spatial correlation of HSI by learning local linear spatial propagation, thereby maintaining the spatial consistency of HSI and further optimizing the classification results. Xiong et al. [51] proposed a novel attention initiation module that learns more representative spatial features using convolutional filters of different spatial sizes and a dynamic CNN architecture, thereby improving classification accuracy. Hou et al. [52] proposed a novel contextual information and spatial attention-based network (CANet) that integrates background information and a spatial attention mechanism. This network encodes long-range spatial dependencies through CA blocks, combines spatial attention encoders, and decodes context information into channel-specific biases, thereby efficiently extracting rich contextual semantic information. In addition, Chen et al. [53] proposed a global contextual spatial attention network, which embeds the global contextual spatial attention mechanism into a densely connected backbone network to achieve effective encoding of remote sensing scene image contextual information into local features. While the methods described above are effective for spatial context aggregation, they lack explicit coordinate encoding and head decoupling, which may overlook precise geometric relationships between pixels and produce entangled feature representations between different attention heads. This limitation weakens the model's ability to capture long-range spatial dependencies while preserving local geometry.

C. Dual-Branch Architectures and Spectral–Spatial Feature Fusion Strategies

In recent years, the dual-branch architecture has received increasing attention, as it effectively extracts rich spatial and spectral information from HSI by extracting features independently in the spectral and spatial dimensions. For instance, Kan et al. [54] designed a multiscale separable convolutional feature extraction method, which uses a parallel structure to extract multiscale spatial and spectral features, thereby achieving efficient mining of spectral–spatial features. Shi et al. [55] proposed a dual-branch multiscale transformer model, which extracts rich discriminative features at different scales, makes full use of the multiscale spatial background and spectral characteristics of HSI, and thus significantly improves its classification performance. Ashraf et al. [56] proposed a spatial–spectral bidirectional encoder representation from transformers (BERTs) model for HSI classification, which takes into account both spatial correlation and spectral correlation, effectively overcoming the limitation of the original BERT algorithm that only focuses on spatial information. To fully exploit the complementary nature of spectral and spatial information, numerous studies have investigated spectral–spatial fusion strategies for HSI classification. Li et al. [57] proposed a spectral–spatial feature learning (SSFL) method based on a multifeature fusion strategy and a deep hierarchical architecture. This method achieves effective extraction of spectral–spatial features through SSFL and fuses the texture features of local binary pattern (LBP) images with the extracted spectral–spatial features. Wan et al. [58] proposed an efficient spectral spatial fusion (ESSF) method. By optimizing the spectral spatial fusion representation and introducing an adaptive attention mechanism to enhance the attention to relevant features, and then enhancing the understanding of the internal context of the image through a context-aware fusion network (CFN), the method achieves efficient fusion of spectral and spatial features to improve classification accuracy. Although such methods explicitly utilize a bi-branch structure to decouple spectral and spatial processing, feature fusion is typically performed in a simple join and summation manner without explicitly modeling cross-branch correspondences or achieving semantic alignment between spectral and spatial representations, which limits the interaction between heterogeneous features.

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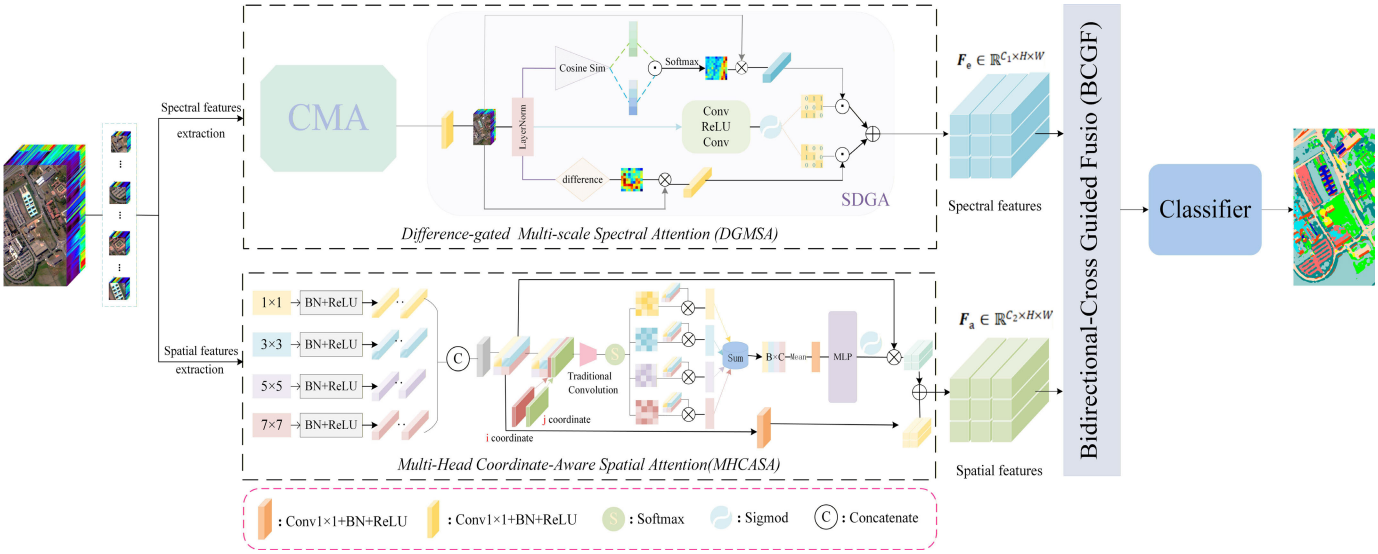


Fig. 1. Proposed MS^{2}CAFNet is composed of three core modules. The DGMSA is used to extract and model spectral features by spectral feature reconstruction and a similarity–difference gated mechanism. The MHCASA module is used to extract and model spatial features, combining multiscale convolution and a coordinate-awareness mechanism. Finally, the fusion module is used to fuse spectral and spatial features using a bidirectional interaction strategy.

III. Methodology

This section provides a detailed description of the implementation of MS^{2}CAFNet, whose overall architecture is illustrated in Fig. 1. The network consists of three core modules, each dedicated to spectral modeling, spatial modeling, and cross-modal feature fusion. In the spectral domain, the multilevel spectral feature reconstruction and similarity-difference gating mechanism are used to jointly model in the DGMSA module, thereby enhancing the discriminability and robustness of spectral feature representations. In the spatial domain, the multiscale convolution and coordinate-aware attention are designed in the MHCASA module to model local neighborhood feature response and cross-regional context information simultaneously. During the feature fusion stage, a bidirectional interaction strategy is employed in the BCGF module to effectively integrate spectral and spatial features, fully leveraging their complementarity to yield more discriminative joint representations, which subsequently facilitates accurate classification. The overall workflow of the proposed MS^{2}CAFNet is summarized in Algorithm 1, which describes the interaction between the feature extraction module, DGMSA, MHCASA, and the BCGF fusion module.

A. Data Preprocessing

In this study, an HSI is denoted as $\boldsymbol{X}_H \in \mathbb{R}^{S\times P\times D}$, where S and P represent the height and width of the image, and D denotes the number of spectral bands. To fully exploit the spatial–spectral contextual information, we adopt a central-pixel extraction strategy. Specifically, from the original HSI, we extract a 3-D cube $\boldsymbol{X} \in \mathbb{R}^{H\times W\times D}$ entered on the target pixel, and construct a sample set $\mathcal{S} = \{x_1, x_2, \ldots, x_i\}$, where H and W indicates the spatial size of each patch. Correspondingly, the label set is denoted as $Y = \{y_1, y_2, \ldots, y_n\}$, where each label $y_i \in \{1, 2, \ldots, K\}$ represents the class to which the sample belongs, and K is the total number of categories. Finally, the class label of the central pixel is used to supervise the training of the model.

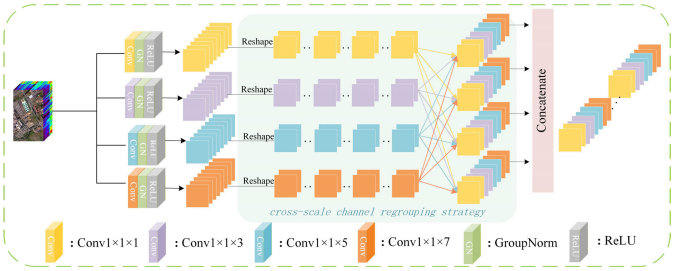


Fig. 2. CMA module is used for the efficient fusion and refinement of multiscale spectral information.

B. Difference-Gated Multiscale Spectral Attention

Although HSIs can finely characterize ground features, they suffer from band redundancy and spectral similarity among neighboring pixels. This makes it difficult for existing methods to capture key discriminative features. To address this, a DGMSA module is proposed, which achieves fine and robust spectral representation through multilevel spectral feature reconstruction and a similarity-difference gating mechanism. Its detailed structure is shown in Fig. 1. It includes two modules: the channel-aware multiscale aggregation (CMA) module and the similarity-difference gated attention (SDGA) module. Within DGMSA, CMA and SDGA play complementary yet sequential roles. CMA is responsible for cross-scale spectral interaction, where channel responses extracted from different receptive fields are reorganized and aggregated to enhance multiscale spectral dependency modeling. SDGA is subsequently applied to the aggregated features to perform similarity–difference adaptive weighting.

The detailed structure of CMA is illustrated in Fig. 2. This module is designed with four parallel 3-D convolutional

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Algorithm 1 Pseudo-Code of the Proposed MS^{2}CAFNet

Require: Input: HSI $\boldsymbol{X}_H \in \mathbb{R}^{S\times P\times D}$
Output: Y predicted class labels
Network Hyper-parameters: Adam optimizer, Learning Rate = 5e-4(PU), Learning Rate = 5e-4(Houston), Learning Rate = 5e-4(IP), Learning Rate = 3e-4(Honghu); Batch Size = 64; training sample rate = 1%(PU), Houston, Honghu; training sample rate = 1%(IP);
Ensure: Ensemble classifier outputs for each pixel;

$\boldsymbol{X} \leftarrow$ preprocess original HSI into cube $\boldsymbol{X}_H$
Multi-scale spectral-spatial feature extraction for HSI
Difference-Gated Multi-scale Spectral Attention (DGMSA) module
Channel-Aware Multi-Scale Aggregation (CMA)
$\boldsymbol{F}_p$, $\boldsymbol{F}_s$, $\boldsymbol{F}_m$ and $\boldsymbol{F}_l \leftarrow$ CMA($\boldsymbol{X}$)
$\boldsymbol{F}_{CMA} \leftarrow$ Apply cross-scale channel regrouping strategy on $\boldsymbol{F}_p$, $\boldsymbol{F}_s$, $\boldsymbol{F}_m$ and $\boldsymbol{F}_l$
Similarity-Difference Gated Attention (SDGA)
$\boldsymbol{F}_{smi} \leftarrow$ Similarity-guided attention branch ($\boldsymbol{F}_{CMA}$)
$\boldsymbol{F}_{diff} \leftarrow$ Similarity-guided attention branch ($\boldsymbol{F}_{CMA}$)

$$\boldsymbol{F}_e = \boldsymbol{G} \odot \boldsymbol{F}_{sim} + (1-\boldsymbol{G}) \odot \boldsymbol{F}_{diff} \in \mathbb{R}^{B\times C_1\times H\times W}$$

Multi-Head Coordinate-Aware Spatial Attention (MHCASA)
$\boldsymbol{F} \leftarrow$ Multi-scale spatial feature extraction ($\boldsymbol{X}$)
$\boldsymbol{F}_g \leftarrow$ Global information aggregation pathways ($\boldsymbol{X}$)
$\boldsymbol{F}_l \leftarrow$ local convolutional pathways($\boldsymbol{F}$)

$$\boldsymbol{F}_a = \text{Dropout}\left(\boldsymbol{F}_g + \boldsymbol{F}_l\right) \in \mathbb{R}^{B\times C_2\times H\times W}$$

Bidirectional Cross-Guided Fusion (BCGF)
$\boldsymbol{F}_{fused} \leftarrow$ Align and fuse of spectral and spatial features ($\boldsymbol{F}_e$, $\boldsymbol{F}_a$)
Classification Module
Out $\leftarrow$ Linear(Droput(GeLU(LayerNorm(Linear (Avg($\boldsymbol{F}_{\text{fused}}$))))))
Update Trainable Parameters
Loss $\leftarrow$ CrossEntropy(Out)
Backpropagation + Parameter update
return Y

branches, whose kernel sizes along the spectral dimension are set to 3, 5, 7, and 9, respectively. By preserving the spatial resolution, CMA can effectively capture multiscale spectral features, thereby enhancing the feature representation capacity from fine-grained to coarse-grained levels. The outputs of each branch are further processed with group normalization and nonlinear activation, which improves the stability of feature distributions while retaining channel-specific information. The corresponding formulation is as follows:

$$\boldsymbol{F}_i = \text{ReLU}\left(\text{GN}\left(\text{Conv3D}_{(1,1,k_i)}(\boldsymbol{X})\right)\right), \quad i \in \{p,s,m,l\} \tag{1}$$

where $\text{Conv3D}_{(1,1,k_i)}(\cdot)$ represents the 3-D convolution operation with a kernel size of k_i along the spectral dimension, where $i \in \{p,s,m,l\}$ corresponds to the four parallel branches. Where p, s, m, and l denote the four convolution branches with different receptive fields, corresponding to kernel sizes of 3×3, 5×5, 7×7, and 9×9, respectively. These branches are designed to capture spectral–spatial features at multiple scales.

To enhance cross-scale spectral interaction, we introduce a cross-scale channel regrouping strategy, which reorganizes channel responses generated at different receptive-field scales into complementary channel groups. Instead of directly aggregating multiscale features, the regrouping operation redistributes channel responses to model interscale spectral dependencies and reduce redundancy. Specifically, feature maps at each scale are grouped and reconstructed along the channel dimension, dividing them into multiple feature subsets. Subsequently, according to the grouping order, the corresponding subfeatures from different scales are concatenated across scales, enabling more sufficient fusion of fine- and coarse-grained information. Specifically, the multiscale feature maps $\boldsymbol{F}_p$, $\boldsymbol{F}_s$, $\boldsymbol{F}_m$, and $\boldsymbol{F}_l$ are divided into multiple subsets based on a predefined group number g, and cross-scale concatenation is performed within each group to obtain multigranularity fused features $\boldsymbol{F}_{psml}^{g_j}$. The computation of the cross-scale channel regrouping strategy is formulated as follows:

$$\boldsymbol{F}_i^{g_j} = \text{Reshape}\left(\boldsymbol{F}_i\right) \in \mathbb{R}^{g,H\times W\times D} \tag{2}$$

$$\text{Group}\left(\boldsymbol{F}_i, g\right) = \left[\boldsymbol{F}_i^{g_1}, \boldsymbol{F}_i^{g_2}, \ldots, \boldsymbol{F}_i^{g_j}\right], \quad i \in \{p,s,m,l\} \tag{3}$$

$$\boldsymbol{F}_{psml}^{g_j} = \text{Concat}\left(\boldsymbol{F}_p^{g_j}, \boldsymbol{F}_s^{g_j}, \boldsymbol{F}_m^{g_j}\right) \tag{4}$$

where $j \in \{1,2,\ldots,g\}$ denotes the index of the channel group. Subsequently, all the group-wise fused features are concatenated across groups to form the final integrated representation $\boldsymbol{F}_{psml}$

$$\boldsymbol{F}_{psml} = \text{Concat}_{j=1}^{g}\left(\boldsymbol{F}_{psml}^{g_j}\right) \tag{5}$$

$$\boldsymbol{F}_{CMA} = \text{Reshape}\left(\boldsymbol{F}_{psml}\right) \in \mathbb{R}^{H\times W\times D}. \tag{6}$$

Finally, the CMA module generates the output feature $\boldsymbol{F}_{CMA}$ by effectively fusing and refining multiscale information, resulting in more structured and discriminative spectral representations.

To simultaneously ensure the stability and discriminability of HSI features, the SDGA module is proposed, whose detailed structure is illustrated in Fig. 1. Specifically, the features generated by CMA are first fed into SDGA for channel normalization. Subsequently, a cosine similarity matrix across spectral bands is computed to characterize spectral correlations, which serves to highlight the most informative spectral bands. The corresponding formulation is as follows:

$$\boldsymbol{v}_i = \text{Reshape}\left(\text{LN}_{\text{channel}}\left(\boldsymbol{F}_{CMA}\right)\right) \in \mathbb{R}^{ND},\ N = H\times W \tag{7}$$

$$\boldsymbol{v}_i^{\text{norm}} = \frac{\boldsymbol{v}_i}{\|\boldsymbol{v}_i\|_2} \tag{8}$$

$$\boldsymbol{S}_{ij} = \boldsymbol{v}_i^{norm} \cdot \boldsymbol{v}_j^{norm} \tag{9}$$

where $\boldsymbol{v}_i$ denotes the original spectral band vector, D represents the number of spectral bands, $\|\boldsymbol{v}_i\|_2$ is the L2 norm of the vector, $\boldsymbol{v}_i^{norm}$ denotes the normalized vector, and $\boldsymbol{v}_j^{norm}$ represents the transpose of $\boldsymbol{v}_i^{\text{norm}}$, and $\boldsymbol{S}_{ij}$ represents the similarity matrix. Next, the similarity matrix is normalized via a softmax function to produce spectral similarity attention scores. These

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scores guide a weighted aggregation of the feature $\boldsymbol{F}_{CMA}$ to generate the similarity feature $\boldsymbol{F}_{sim}$. This operation strengthens common spectral responses while suppressing noise and redundant information, thus improving intraclass consistency and representation robustness. The above process can be formally expressed as

$$\tilde{\boldsymbol{S}}_{ij} = \frac{exp\left(\boldsymbol{S}_{ij}\right)}{\sum_{k=1}^{N} exp\left(\boldsymbol{S}_{ik}\right)} \tag{10}$$

$$\boldsymbol{F}_{sim} = \text{Reshape}\left(\sum_{j=1}^{N} \tilde{\boldsymbol{S}}_{ij} \cdot \boldsymbol{F}_{CMA}\right) \tag{11}$$

where $\tilde{\boldsymbol{S}}_{ij}$ denotes the similarity between the ith and jth positions.

However, relying solely on spectral similarity for classification may lead to over-smoothing or homogeneity in the learned features, thereby obscuring a few highly discriminative spectral bands. To address this issue, a different branch is further designed to effectively capture these complementary fine-grained features, alleviating the adverse effects of over-smoothing. Specifically, we define the difference score as

$$\boldsymbol{d}_{i,j} = 1 - \frac{x_i \cdot x_j}{\|x_i\|_2 \|x_j\|_2} \tag{12}$$

where x_i and x_j denote the spectral band vectors at the ith and jth positions, respectively. Although cosine similarity captures the global angular consistency between spectral vectors, it mainly reflects the overall orientation of spectra and is insensitive to localized band-wise deviations. The difference score $\boldsymbol{d}_{i,j}$ measures the degree of dissimilarity between spectral bands, with larger values indicating more significant discrepancies. In contrast, the difference operation explicitly measures the magnitude of spectral variation between corresponding bands, which enables the model to detect abrupt spectral changes. Such variations are associated with boundary-sensitive spectral bands critical for distinguishing neighboring classes. Therefore, the difference term provides complementary information to cosine similarity rather than being redundant. Based on this score, the low-correlation bands of the features $\boldsymbol{F}_{CMA}$ are aggregated to generate difference-aware attention features $\boldsymbol{F}_{diff}$, which emphasize discrepant regions along the spectral curves that are often more directly indicative of interclass separability. The overall process is formulated as

$$\boldsymbol{F}_{diff} = \text{Reshape}\left(\sum_{j=1}^{N} \boldsymbol{d}_{ij} \cdot \boldsymbol{F}_{CMA}\right). \tag{13}$$

To reduce the intraclass variability of spectral representations while enlarging the interclass distances, a simple gating mechanism is employed to adaptively balance similarity and difference features on a position-wise basis. Specifically, a structure consisting of two $1 \times 1 \times 1$ convolutional layers with a reduction ratio and a nonlinear activation, which is adopted to learn a dynamic gating map. The gating operation aggregates the features and normalizes them into the range [0, 1] via a sigmoid activation function, thereby enabling voxel-wise modulation of the fusion between similarity-enhanced and difference-enhanced representations. Formally, the fused representation is computed as

$$\boldsymbol{F}_e = \boldsymbol{G} \odot \boldsymbol{F}_{sim} + (1 - \boldsymbol{G}) \odot \boldsymbol{F}_{diff} \tag{14}$$

where $\boldsymbol{G}$ denotes the learnable dynamic weight map, and $\odot$ represents element-wise multiplication. This design enables the network to favor robust similarity aggregation in homogeneous regions, while emphasizing complementary difference cues in boundaries or mixed pixels, thereby better aligning with the discriminative objective of intraclass compactness and interclass separability.

C. Multihead Coordinate-Aware Spatial Attention

Spectral modeling provides the foundation for HSI classification, while spatial features, carrying information such as texture and shape, are a critical supplement for distinguishing similar targets in complex scenes. However, traditional methods have shortcomings in global context capture, local detail preservation, and location information modeling. Therefore, the MHCASA module is designed tos balance global modeling capabilities with local details to improve the representation of complex spatial structures.

First, the input hyperspectral cube $\boldsymbol{X} \in \mathbb{R}^{S \times S \times D}$ is reshaped for the 2-D convolutional processing. It is then fed into four parallel 2-D convolutional branches with different receptive fields to extract hierarchical spatial semantic features. The resulting four groups of features are concatenated along the channel dimension and subsequently compressed by a 1×1 convolution, which also facilitates cross-scale feature fusion. This process produces a spatially fused feature map with hierarchical structure-awareness. The formulation is expressed as follows:

$$\text{Conv}_{2D}^{k\times k}(\boldsymbol{X}) = \text{ReLU}\left(\text{BN}\left(\boldsymbol{X} * \boldsymbol{\omega}^T + b\right)\right),\ k \in \{1,3,5,7\} \tag{15}$$

$$\boldsymbol{f}_n = \text{Conv}_{2D}^{k\times k}(\boldsymbol{X}),\ n \in \{1,2,3,4\} \tag{16}$$

$$\boldsymbol{F} = \text{Concate}\left(\boldsymbol{f}_1, \boldsymbol{f}_2, \boldsymbol{f}_3, \boldsymbol{f}_4\right) \tag{17}$$

where $\text{Conv}_{2D}^{k\times k}(\boldsymbol{X})$ denotes a 2-D convolutional block, $*$ indicates the convolution operation, $\boldsymbol{\omega}^T$ stands for the convolutional kernel weights, b is the bias term, and $\boldsymbol{f}_1$, $\boldsymbol{f}_2$, $\boldsymbol{f}_3$, and $\boldsymbol{f}_4$ correspond to the outputs of the 2-D convolutional blocks with different receptive fields, respectively.

After obtaining the multiscale fused feature map, a multihead global context aggregation attention mechanism is employed to capture long-range dependencies. Specifically, to enable the model to perceive absolute spatial positional information when handling HSI in the case of complex geometric structures or uneven target distributions, we incorporate a CoordConv into MHCASA. In particular, CoordConv explicitly encodes spatial positional information by concatenating normalized coordinate channels to the original feature map along the channel dimension. This process can be formally expressed as

$$\boldsymbol{P}_{i(h,w)} = \frac{2h}{H-1} - 1, \quad \boldsymbol{P}_{j(h,w)} = \frac{2w}{W-1} - 1 \tag{18}$$

where (h, w) denotes the spatial position of a pixel, while $\boldsymbol{P}_i \in \mathbb{R}^{H\times W}$ and $\boldsymbol{P}_j \in \mathbb{R}^{H\times W}$ correspond to its vertical and

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horizontal coordinate mappings, respectively. These coordinate maps are expanded and concatenated with the input features, after which a standard 2-D convolution operation is applied to extract position-aware representations

$$\boldsymbol{F}_{coord} = \text{Concat}\left(\boldsymbol{F}, \boldsymbol{P}_i, \boldsymbol{P}_j\right) \tag{19}$$

$$\boldsymbol{F}_{b,o,h,w} = \sum_{c=1}^{C+2}\sum_{u=1}^{k_H}\sum_{v=1}^{k_W} \boldsymbol{W}_{o,c,u,v} \cdot \boldsymbol{F}_{b,c,h+m,w+n} + b_o \tag{20}$$

where $\boldsymbol{F}_{b,o,h,w}$ is the feature value at spatial position (h, w), for the oth output channel of the bth sample in the output feature map. C is the number of channels of the input feature map. k_H and k_W represent the height and width of the convolution kernel, respectively. $\boldsymbol{W}_{o,c,u,v}$ represent the weights of the convolution kernel, where o corresponds to the output channel, C corresponds to the input channel, and m and n correspond to the spatial positions within the convolutional kernel. b_o denotes the convolutional kernel weights and bias term.

Subsequently, a softmax normalization is applied along the spatial dimension to generate multihead attention masks, enabling the model to differentiate the weight distribution across different spatial regions during long-range dependency modeling. Thereafter, each attention head independently performs weighted aggregation on the input features, thereby capturing multiview global contextual representations. The process can be formally expressed as

$$\boldsymbol{M} = \text{Conv}_{1\times1}\left(\boldsymbol{F}_{b,o,h,w}\right) \tag{21}$$

$$\boldsymbol{g} = \frac{1}{H_d}\sum_{i=1}^{H_d}\left(\sum_{h=1}^{H}\sum_{w=1}^{W} \boldsymbol{x}_{b,:,h,w} \cdot \boldsymbol{M}^{(i)}_{b,1,h,w}\right) \tag{22}$$

where $\boldsymbol{g}$ denotes the final contextual vector obtained by averaging across all heads, which H_d represents the number of attention heads, and $\boldsymbol{x}_{b,:,h,w}$ represents the channel-wise vector of the bth sample at spatial position (h, w), while $\boldsymbol{M}^{(i)}$ denotes the corresponding attention weights. To effectively enhance semantically relevant global features, the aggregated global contextual vector is passed through a two-layer feed-forward network, followed by a sigmoid activation to generate a channel-wise attention weight matrix $\boldsymbol{S} \in \mathbb{R}^{B\times C}$. This matrix is then broadcast into a 4-D tensor $\boldsymbol{S}' \in \mathbb{R}^{B\times C\times1\times1}$ to reweight the input features accordingly

$$\boldsymbol{S} = \sigma\left(\text{LN}_2\left(W_2 \cdot \phi\left(\text{LN}_1\left(W_1 \cdot \boldsymbol{g} + b_1\right)\right) + b_2\right)\right). \tag{23}$$

$$\boldsymbol{F}_g = \boldsymbol{F} \cdot \boldsymbol{S}'. \tag{24}$$

To compensate for the potential loss of local information induced by global modulation, a depthwise separable convolution is incorporated in parallel to preserve fine-grained spatial cues such as edges and textures. Subsequently, the local convolutional features are integrated with the global contextual representations through element-wise addition, and a Dropout layer is applied at the output stage to alleviate the risk of overfitting. Specifically, as shown in the formula

$$\boldsymbol{F}_l = \text{ReLU}\left(\text{BN}\left(\text{DWConv}_{3\times3}\left(\boldsymbol{F}\right)\right)\right) \tag{25}$$

$$\boldsymbol{F}_a = \text{Dropout}\left(\boldsymbol{F}_g + \boldsymbol{F}_l\right). \tag{26}$$

Therefore, the MHCASA module can simultaneously model multiscale spatial features, global dependencies, and local details, thereby obtaining a more discriminative spatial representation.

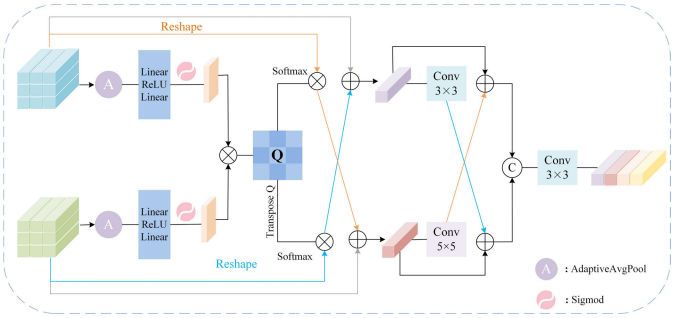


Fig. 3. Core of BCGF is a bidirectional cross-interaction mechanism.

D. Bidirectional Cross-Guided Fusion

In HSIs, spectral features excel at material identification but have weak noise resistance, while spatial features possess strong geometric representation capabilities but lack sufficient boundary discrimination. Relying solely on a single modality cannot fully leverage the advantages of either. Therefore, exploring their complementarity is crucial for improving classification performance. To this end, the BCGF module is designed to achieve synergistic optimization of spectral and spatial features through a bidirectional interaction mechanism, enhancing feature discriminative power and robustness. Its detailed structure is shown in Fig. 3.

Specifically, the DGMSA and MHCASA modules extract multichannel spectral and spatial feature maps, denoted as $\boldsymbol{F}_e \in \mathbb{R}^{C_1\times H\times W}$ and $\boldsymbol{F}_a \in \mathbb{R}^{C_2\times H\times W}$, respectively. Directly building channel-wise correlation between spectral and spatial features is computationally expensive. It also hardly enhances intrachannel dependencies. To address this issue, in the BCGF module, first global adaptive average pooling is applied to compress spectral and spatial features into representative channel descriptors, each element reflecting the importance of a specific channel

$$\boldsymbol{u} = \text{AAP}\left(\boldsymbol{F}_e\right), \quad \boldsymbol{v} = \text{AAP}\left(\boldsymbol{F}_a\right). \tag{27}$$

Subsequently, a Squeeze-and-Excitation structure is employed to generate channel-wise weights and capture salient single-modal features, enabling the network to adaptively adjust the relative importance of spectral and spatial representations

$$\boldsymbol{F}_{\text{eattn}} = \sigma\left(W_4\delta\left(W_3\boldsymbol{u}\right)\right), \quad \boldsymbol{F}_{\text{aattn}} = \sigma\left(W_6\delta\left(W_5\boldsymbol{v}\right)\right) \tag{28}$$

where and W_3, W_4, W_5, and W_6 denote learnable parameters, $\delta(\cdot)$ represents the GELU activation function, and $\sigma(\cdot)$ denotes the sigmoid function. To explicitly model the dependency between spectral and spatial features, a bidirectional correlation matrix $\boldsymbol{Q} \in \mathbb{R}^{C_1\times C_2}$ is constructed. The matrix $\boldsymbol{Q}$ characterizes the pairwise correlations between spectral and spatial channels. Subsequently, softmax normalization is applied along different dimensions

$$\boldsymbol{Q} = \boldsymbol{F}_{\text{eattn}} \otimes \boldsymbol{F}^T_{\text{aattn}} \tag{29}$$

$$\boldsymbol{Q}_s = \text{Softmax}\left(\boldsymbol{Q}\right), \quad \boldsymbol{Q}_t = \text{Softmax}\left(\boldsymbol{Q}^T\right) \tag{30}$$

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where $\otimes$ denotes the outer product operation, which $\boldsymbol{Q}_s$ is used to model the dependency from spectral to spatial features, while $\boldsymbol{Q}_t$ captures the dependency from spatial to spectral features.

To enable bidirectional information exchange between spectral and spatial representations, the correlation matrix is employed for feature projection. Specifically, the input spectral and spatial features are first flattened into matrix forms, denoted as $\boldsymbol{F}_{eflat} \in \mathbb{R}^{C_1\times(H\times W)}$ and $\boldsymbol{F}_{aflat} \in \mathbb{R}^{C_2\times(H\times W)}$, respectively. The spectral features are then projected into the spatial domain, and conversely, the spatial features are projected into the spectral domain. Finally, the projected features are reshaped back to their original dimensions. This process allows spectral representations to complement spatial features with enhanced material discriminability, while spatial representations provide structural priors to reinforce spectral features

$$\boldsymbol{F}'_e = \text{Reshape}\left(\boldsymbol{Q}_s \cdot \boldsymbol{F}_{aflat}\right), \quad \boldsymbol{F}'_a = \text{Reshape}\left(\boldsymbol{Q}_t \cdot \boldsymbol{F}_{eflat}\right). \tag{31}$$

After completing the bidirectional interaction, a residual connection is applied to preserve the original information. This design not only ensures the incorporation of newly learned spectral–spatial interactions but also prevents the loss of the fundamental discriminative information of the original features, thereby enhancing the stability of the overall representation

$$\widehat{\boldsymbol{F}}_a = \boldsymbol{F}_a + \boldsymbol{F}'_a, \quad \widehat{\boldsymbol{F}}_e = \boldsymbol{F}_e + \boldsymbol{F}'_e. \tag{32}$$

In addition to feature fusion at the channel level, the BCGF module also performs feature fusion at the spatial level to achieve better complementarity between spectral and spatial representations. To fully integrate spatial features, convolutional operations with different receptive fields are introduced to further capture local contextual information. Specifically, a 3×3 convolution is first applied to the spectral-enhanced features $\widehat{\boldsymbol{F}}_a$ to map them to the same number of channels as the spatial features $\widehat{\boldsymbol{F}}_e$, followed by element-wise addition. This operation leverages spectral information to guide the update of spatial features, enhancing the material discriminability of spatial representations. Conversely, a 5×5 convolution is applied to the spatial-enhanced features $\widehat{\boldsymbol{F}}_e$ to capture larger local receptive fields, and the result is added to the spectral features $\widehat{\boldsymbol{F}}_a$, enabling spectral representations to absorb more local structural patterns. This process can be formally expressed as

$$\boldsymbol{U} = \text{Conv}_{3\times3}\left(\widehat{\boldsymbol{F}}_a\right) + \widehat{\boldsymbol{F}}_e, \quad \boldsymbol{V} = \text{Conv}_{5\times5}\left(\widehat{\boldsymbol{F}}_e\right) + \widehat{\boldsymbol{F}}_a. \tag{33}$$

Finally, the spectral- and spatial-enhanced features are concatenated along the channel dimension to form a joint representation containing both spectral and spatial information. A subsequent convolution is applied for channel compression and information integration, yielding the final fused output. This process ensures thorough interaction between spectral and spatial features during fusion, producing representations that are both more discriminative and robust, thereby providing stronger feature support for HSI classification.

TABLE I
IS CLASSES AND THE NUMBER OF SAMPLES IN THE PU

No.	Class name	Train	Val	Test
C1	Asphalt	66	66	6499
C2	Meadows	186	186	18277
C3	Gravel	20	20	2059
C4	Trees	30	30	3004
C5	Metal-sheets	13	13	1319
C6	Bare-soil	50	50	4929
C7	Bitumen	13	13	1304
C8	Bricks	36	36	3610
C9	Shadows	9	9	929
/	Total	423	423	41930

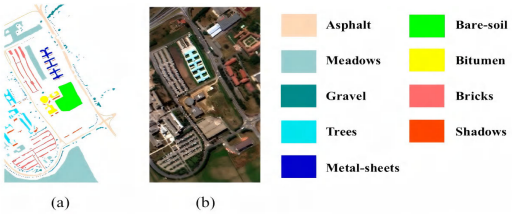


Fig. 4. PU: (a) ground truth and (b) false-color image.

E. Classifier

To perform the final classification, we design a lightweight fully connected classifier composed of two linear layers. Specifically, the fused features processed by the BCGF module are first passed through a linear layer, followed by LayerNorm to stabilize the training process and a GELU activation to enhance nonlinear expressiveness. To mitigate the risk of overfitting, a Dropout regularization is also applied within the classifier. The final linear layer then maps the intermediate features to the target number of classes, producing the ultimate classification predictions.

IV. EXPERIMENT AND ANALYSIS

A. Dataset Description

To evaluate the effectiveness of the proposed MS^{2}CAFNet, eight state-of-the-art DL-based methods are selected for comparison. Extensive experiments are conducted on the Pavia University (PU), Indian Pines (IP), Houston University (Houston), and WHU-Hi-HongHu (HongHu) datasets to demonstrate the superiority and generalization capability of MS^{2}CAFNet in HSI classification tasks. The detailed descriptions of these four datasets are provided as follows.

1) *PU:* This dataset was acquired by the ROSIS sensor and captures the campus of the University of Pavia and its surrounding urban areas in northern Italy. The image has a spatial size of 610×340 pixels, covering wavelengths from 0.43 to 0.86 μm with a spatial resolution of 1.3 m/pixel. In our experiments, 103 effective spectral bands are retained after removing noisy bands for classification. As summarized in Table I, 1%, 1%, and 98% of the samples are randomly selected for training, validation, and testing, respectively. The false-color composite image and the corresponding ground truth map are illustrated in Fig. 4.

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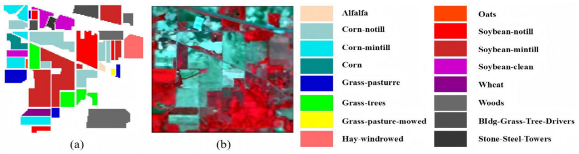


Fig. 5. IP: (a) ground truth and (b) false-color image.

TABLE II

IS CLASSES AND THE NUMBER OF SAMPLES IN THE IP

No.	Class name	Train	Val	Test
C1	Alfalfa	2	2	42
C2	Corn-notill	71	71	1286
C3	Corn-mintill	41	41	748
C4	Corn	11	11	215
C5	Grass-pasture	24	24	435
C6	Grass-trees	36	36	658
C7	Grass-pasture-mowed	1	1	26
C8	Hay-windrowed	23	23	432
C9	Oats	1	1	18
C10	Soybean-notill	48	48	876
C11	Soybean-mintill	122	122	2211
C12	Soybean-clean	29	29	535
C13	Wheat	10	10	185
C14	Woods	63	63	1139
C15	BIdg-Grass-Tree-Drivers	19	19	348
C16	Stone-Steel-Towers	4	4	85
/	Total	505	505	9239

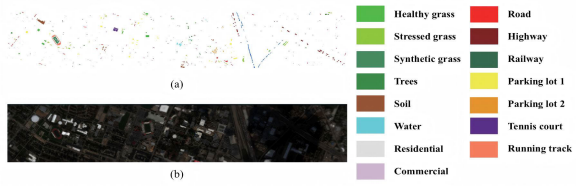


Fig. 6. Houston: (a) ground truth and (b) false-color image.

2) *IP:* This dataset was collected in 1992 by the AVIRIS sensor of NASA over agricultural areas near Purdue University, Indiana, USA. The scene consists of 145 × 145 pixels with a spatial resolution of 20 m/pixel, covering 224 spectral bands spanning the 0.4–2.5-μm wavelength range. After removing bands affected by water vapor absorption, 200 effective spectral bands are retained for our experiments. The dataset contains 16 land-cover classes, primarily including corn, soybean, grass, forest, and built-up areas, representing both agricultural and nonagricultural surfaces. In our experiments, 5%, 5%, and 90% of the labeled samples are randomly selected for training, validation, and testing, respectively, as detailed in Table II. The false-color composite image and the corresponding ground truth map are shown in Fig. 5.
3) *Houston:* This dataset was collected on June 23, 2012, by the National Center for Airborne Laser Mapping (NCALM), funded by the U.S. National Science Foundation (NSF), over the campus of the University of Houston. The image has spatial dimensions of 349 × 1905 pixels with a spatial resolution of approximately 2.5 m/pixel, covering 144 spectral bands within the wavelength range of 380–1050 nm. Due to computational constraints, only 30 representative spectral bands are utilized in our experiments. The dataset consists of 15 land-cover classes. A total of 1%, 1%, and 98% of the samples are randomly selected for the training, validation, and testing sets, respectively, as detailed in Table III. The false-color composite image and its corresponding ground truth map are presented in Fig. 6.
4) *HongHu:* This dataset was acquired by Wuhan University using a Headwall Nano-Hyperspec sensor mounted on an unmanned aerial vehicle (UAV) over a complex agricultural region in Honghu, Hubei Province, China. The image has spatial dimensions of 940 × 475 pixels with an approximate spatial resolution of 0.043 m/pixel, and it comprises 270 spectral bands spanning the

TABLE III

IS CLASSES AND THE NUMBER OF SAMPLES IN THE HOUSTON

No.	Class name	Train	Val	Test
C1	Healthy grass	12	12	1227
C2	Stressed grass	12	12	1230
C3	Synthetic grass	6	6	685
C4	Trees	12	12	1220
C5	Soil	12	12	1218
C6	Water	3	3	319
C7	Residential	12	12	1244
C8	Commercial	12	12	1220
C9	Road	12	12	1228
C10	Highway	12	12	1203
C11	Railway	12	12	1211
C12	Parking lot 1	12	12	1209
C13	Parking lot 2	4	4	461
C14	Tennis court	4	4	420
C15	Running track	6	6	648
/	Total	143	143	14743

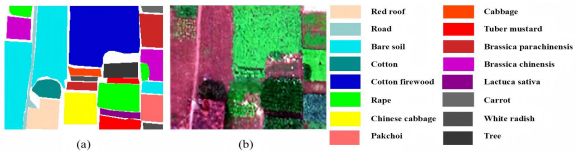


Fig. 7. Honghu: (a) ground truth and (b) false-color image.

TABLE IV

IS CLASSES AND THE NUMBER OF SAMPLES IN THE HONGHU

No.	Class name	Train	Val	Test
C1	Red roof	33	33	3254
C2	Road	14	14	1454
C33	Bare soil	187	187	18351
C4	Cotton	17	17	1758
C5	Cotton firewood	149	149	14641
C6	Rape	58	58	5692
C7	Chinese cabbage	40	40	3974
C8	Pakchoi	23	23	2329
C9	Cabbage	9	9	921
C10	Tuber mustard	25	25	2534
C11	Brassica parachinensis	39	39	3901
C12	Brassica chinensis	43	43	4221
C13	Lactuca sativa	10	10	982
C14	Carrot	5	5	553
C15	White radish	9	9	955
C16	Tree	20	20	1997
/	Total	681	681	67517

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400–1000-nm wavelength range. The scene primarily covers farmland, roads, buildings, and trees, with 22 annotated land-cover classes in total. In our experiments, 16 major categories are selected for classification analysis. In the experiment, 1%, 1%, and 98% of the samples are randomly selected for training, validation, and testing, as detailed in Table IV. The false-color composite image and the corresponding ground truth map are illustrated in Fig. 7.

B. Experiment Setup, Evaluation Metrics, and Comparison Algorithms

1) *Configuration:* All experiments are conducted on a compact DL workstation configured with CUDA v11.6, PyTorch 1.13, and Python 3.8. The tests are performed on a laptop equipped with an Intel[1] Core[2] i5-1035G1 CPU @ 1.00 GHz, 16 GB of RAM, and an NVIDIA GeForce RTX 2080 Ti GPU with 11 GB of VRAM. The proposed MS^{2}CAFNet was trained using the Adam optimizer with an initial learning rate of 0.0005. The batch size was set to 64, and the model was trained for 200 epochs. The learning rate was decayed by a factor of 0.1 every 80 epochs. Cross-entropy loss was used as the optimization objective. The training ratios of PU (1%), IP (5%), and Houston (1%) used in the experiments followed evaluation protocols commonly used in the HSI classification literature. These settings are widely used to simulate few-shot learning scenarios and ensure fair comparisons with existing methods. Because different datasets have different class distributions and sample sizes, dataset-specific training ratios were used to maintain experimental fairness and reproducibility, demonstrating the advantages and versatility of MS^{2}CAFNet in HSI classification tasks.
2) *Evaluation Metrics:* To comprehensively assess the classification performance, three widely used quantitative metrics are adopted. Overall accuracy (OA) measures the overall correctness of classification results. Average accuracy (AA) evaluates the classification consistency across different land-cover categories. Kappa Coefficient (Kappa $\times$ 100) quantifies the agreement between predicted and ground-truth labels beyond chance level.
3) *Comparison Methods:* To thoroughly validate the effectiveness of the proposed MS^{2}CAFNet, eight state-of-the-art methods for HSI classification are selected for comparison, including MCFN [59], DBCTNet [60], DBMST [55], DSNet [48], MASSFormer [61], CTANet [62], DSFormer [63], and MSFICNet [64]. The performance of these methods is compared with that of MS^{2}CAFNet to verify its superiority. The comparative methods can be broadly categorized into three groups: CNN and attention-based models, CNN-Transformer hybrid models, and Transformer-based models. CNN-based approaches (e.g., MCFN and DSNet) primarily rely on multiscale convolutional operations and contextual attention mechanisms to extract spectral–spatial features from limited hyperspectral samples. CNN-Transformer hybrid models (e.g., DBCTNet and CTANet) combine the local feature extraction capability of CNNs with the global modeling strength of Transformers, thereby achieving complementary spectral–spatial feature fusion. Transformer-based models (e.g., DBMST, MASSFormer, DSFormer, and MSFICNet) focus on token-level representation learning and multiscale feature fusion. These methods often incorporate advanced modules such as memory mechanisms, sparse interactions, or adaptive token selection to enhance global dependency modeling and improve classification robustness.

[1]Registered trademark.
[2]Trademarked.

C. Experimental Results and Analysis

To validate the classification performance of the proposed network, experiments are conducted on four publicly available hyperspectral datasets. The proposed MS^{2}CAFNet is compared with eight state-of-the-art methods under the same experimental settings. Independent data partitions are employed for the training, validation, and testing sets, with training samples selected through random sampling. Each experiment is independently repeated five times, and the average results are reported to ensure the reliability and statistical robustness of the findings. The classification results of different methods on the four datasets are summarized in Tables V–VIII, where the best results are highlighted in bold, and the second-best results are underlined. The corresponding full-pixel classification maps for all compared methods are presented in Figs. 8–11.

1) *Experimental Results Analysis on the PU Dataset:* To evaluate the effectiveness of the proposed MS^{2}CAFNet, comparative experiments are conducted on the PU dataset against eight state-of-the-art methods. As shown in Table V, although most methods perform well due to the regular spectral and spatial patterns of the PU dataset, MS^{2}CAFNet still achieves the best overall results. It obtains the highest OA (98.65%) and Kappa $\times$ 100 (98.21%), while its AA (97.95%) is also among the top, outperforming DSNet, DBCTNet, and DSFormer. Regarding class-wise performance, MS^{2}CAFNet consistently keeps superior results. For instance, it achieves 99.54% in C4, which contains complex spatial structures, clearly surpassing other competitors. It also achieves 99.96% accuracy on C6, higher than the second-best method, DBCTNet (99.37%). This demonstrates that MS^{2}CAFNet has a stronger ability to represent spectral features. Even for challenging categories such as C8, prone to misclassification, the network achieves 95.62%, exceeding CTANet (94.78%). As illustrated in Fig. 8, MS^{2}CAFNet generates sharper class boundaries and fewer misclassified regions, indicating its enhanced ability to capture fine spectral–spatial details and reduce category confusion.
2) *Experimental Results Analysis on the IP Dataset:* It contains 16 land-cover classes (including crops, vegetation, and man-made structures) with an extremely imbalanced sample distribution. As summarized in Table VI, MS^{2}CAFNet achieves the best results with an OA

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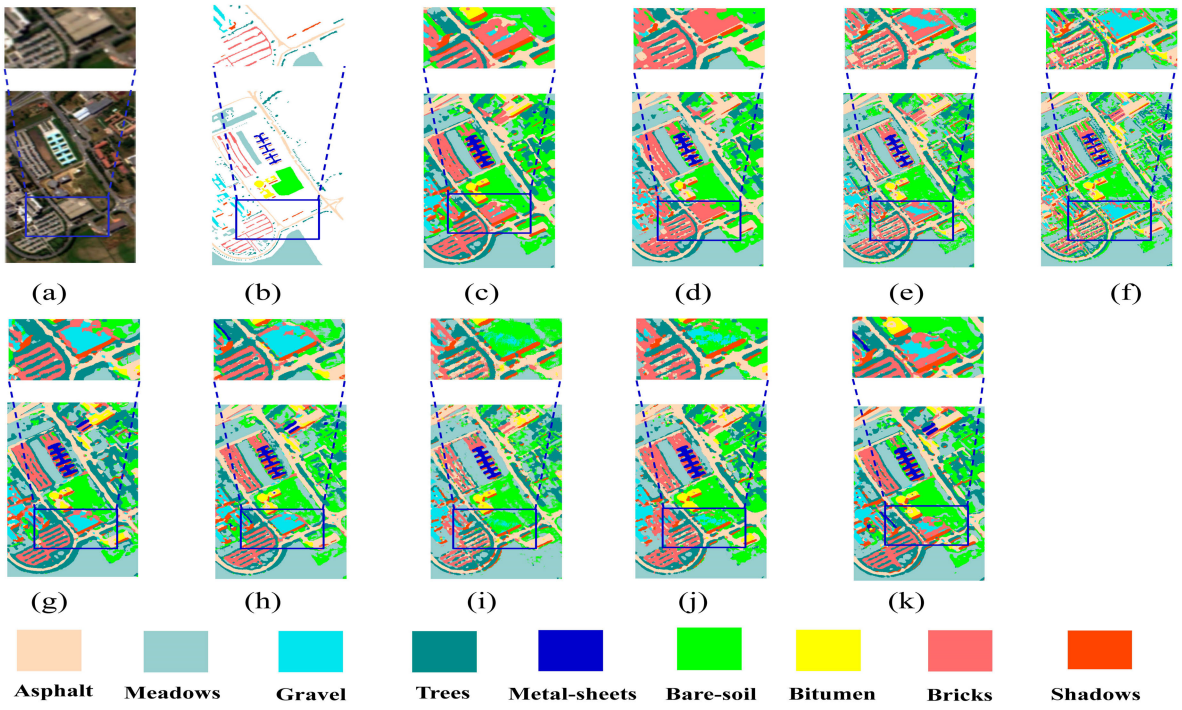


Fig. 8. Full-pixel classification maps for the PU dataset. (a) Pseudo-color image; (b) ground-truth map; (c) MCFN; (d) DBCTNet; (e) DBMST; (f) DSNet; (g) MASSFormer; (h) CTANet; (i) DSformer; (j) MSFICNet; and (k) proposed.

TABLE V
CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA × 100 WITH STANDARD DEVIATION FOR THE PU DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

NO.	MCFN	DBCTNet	DBMST	DSNet	MASSFormer	CTANet	DSformer	MSFICNet	Proposed
C1	**99.25±0.17**	82.43±0.42	96.90±0.42	92.71±0.03	96.36±0.02	95.56±0.30	91.29±0.13	92.68±0.26	99.18±0.05
C2	99.46±0.15	99.56±0.04	98.64±0.05	96.46±0.29	99.56±0.00	99.02±0.04	96.92±0.07	**99.59±0.01**	99.10±0.10
C3	**98.56±0.40**	96.77±0.06	91.07±0.75	88.11±0.18	97.42±0.25	94.95±0.21	77.51±0.24	94.58±0.29	98.62±0.07
C4	**99.82±0.00**	98.02±0.07	99.49±0.04	99.12±0.05	98.18±0.02	96.34±0.25	96.39±0.03	97.02±0.76	99.54±0.05
C5	**99.98±0.03**	92.88±0.21	99.32±0.18	99.77±0.00	99.11±0.02	99.65±0.03	99.60±0.03	99.54±0.00	98.86±0.34
C6	97.25±0.08	99.37±0.06	98.37±0.17	97.06±0.17	98.42±0.01	98.81±0.05	95.09±0.10	98.70±0.15	**99.96±0.07**
C7	92.98±3.85	80.00±4.00	94.16±3.49	90.45±0.14	99.66±0.19	98.79±0.14	98.17±0.17	**100.00±0.00**	95.14±0.55
C8	91.91±1.27	89.78±0.64	89.54±0.69	85.26±0.13	94.56±0.17	94.78±0.15	87.15±0.11	88.11±0.33	**95.62±0.14**
C9	99.82±0.08	99.79±0.15	99.97±0.04	**100.00±0.00**	97.11±0.10	99.44±0.00	99.88±0.00	97.77±0.08	97.36±0.24
OA (%)	98.25±0.30	94.98±0.16	97.14±0.12	94.74±0.12	98.22±0.04	97.72±0.05	94.19±0.03	96.96±0.02	**98.65±0.04**
AA (%)	97.67±0.53	93.18±4.54	96.38±0.40	94.33±0.03	97.82±0.01	97.48±0.04	93.56±0.01	96.44±0.03	**97.95±0.08**
Kappa×100	97.69±0.40	93.30±0.22	96.20±0.16	92.97±0.16	97.63±0.06	96.97±0.07	92.26±0.04	95.97±0.03	**98.21±0.05**

of 97.44%, an AA of 97.26%, and a Kappa × 100 of 97.09%. Compared with the CNN-based DSNet and the hybrid CNN-Transformer model DBMST, MS^{2}CAFNet shows consistent improvements, indicating stronger robustness and generalization when facing heterogeneous and imbalanced data. At the category level, the proposed network achieves 100.00% accuracy in C4 and C7, even under few-sample conditions where other models fail, and attains 99.12% in C15, highlighting its ability to extract both fine spectral details and spatial structural cues. As illustrated in Fig. 9, MS^{2}CAFNet produces more coherent spatial regions and clearer boundaries, particularly in confusing areas such as C2 and C10. These results confirm the superior spectral–spatial complementarity and enhanced discriminative capability of the network for complex hyperspectral scenes.

3) *Experimental Results Analysis on the Houston Dataset:* To further validate the classification capability of the proposed method in complex urban scenes, experiments are conducted on the Houston dataset, which contains 15 land-cover classes characterized by strong interclass similarity and complex spatial distributions. As shown in Table VII, MS^{2}CAFNet achieves the best

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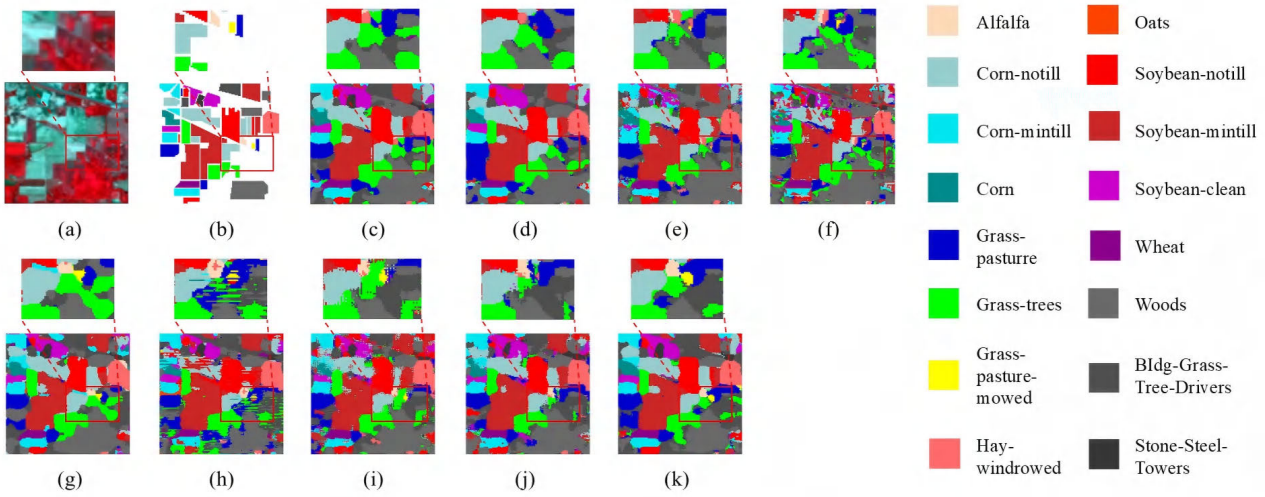


Fig. 9. Full-pixel classification maps for the IP dataset. (a) Pseudo-color image; (b) ground-truth map; (c) MCFN; (d) DBCTNet; (e) DBMST; (f) DSNet; (g) MASSFormer; (h) CTANet; (i) DSformer; (j) MSFICNet; and (k) proposed.

TABLE VI

CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA × 100 WITH STANDARD DEVIATION FOR THE IP DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD, SECOND-BEST UNDERLINED)

NO.	MCFN	DBCTNet	DBMST	DSNet	MASSFormer	CTANet	DSformer	MSFICNet	Proposed
C1	70.18±2.07	80.00±4.00	93.22±0.33	**100.00±0.00**	**100.00±0.00**	92.28±1.35	**100.00±0.00**	**100.00±0.00**	89.33±5.40
C2	96.72±0.26	96.52±0.27	89.07±0.33	92.48±0.02	94.96±0.38	97.07±0.06	93.12±0.25	90.36±0.41	**98.63±0.03**
C3	94.11±0.41	97.10±0.15	91.57±0.08	94.38±0.19	93.79±1.11	96.82±0.20	93.22±0.38	**98.52±0.12**	96.53±0.15
C4	91.46±0.58	98.18±0.18	96.19±0.57	85.99±0.49	94.97±0.25	98.72±0.23	99.29±0.22	97.09±0.05	**100.00±0.00**
C5	**99.33±0.11**	94.91±0.80	96.60±0.33	95.18±0.09	97.26±0.11	93.62±0.18	98.02±0.10	97.48±0.35	99.13±0.01
C6	99.72±0.55	**99.87±0.15**	95.18±0.10	94.94±0.11	94.54±0.40	98.81±0.18	95.73±0.34	97.27±0.11	99.23±0.01
C7	57.74±3.73	0.00±0.00	73.33±3.33	56.94±2.77	73.99±1.40	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C8	**99.86±0.27**	94.53±0.28	89.96±0.27	97.68±0.00	99.63±0.18	97.95±0.00	95.27±0.00	96.04±0.20	97.89±0.07
C9	**100.00±0.00**	0.00±0.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	93.75±0.00	**100.00±0.00**
C10	**98.52±0.36**	93.04±0.17	93.68±0.16	93.58±0.52	94.14±0.54	97.94±0.04	93.22±0.20	90.96±0.07	93.31±0.60
C11	98.57±0.14	**98.61±0.07**	97.93±0.14	94.68±0.07	97.30±0.05	97.16±0.05	94.45±0.59	96.19±0.13	96.74±0.03
C12	89.39±0.57	95.16±0.61	93.72±0.77	93.33±0.30	96.25±0.07	88.00±0.39	91.25±0.33	93.15±0.17	**97.92±0.00**
C13	98.93±0.33	99.22±0.56	99.35±0.21	97.86±0.28	96.73±0.00	99.03±0.52	**100.00±0.00**	97.84±0.00	98.58±0.88
C14	97.76±0.13	98.07±0.22	97.94±0.10	95.69±0.12	**99.20±0.03**	98.12±0.04	96.91±0.05	95.14±0.13	99.09±0.08
C15	92.88±0.10	93.54±0.13	88.09±0.44	85.34±0.50	89.14±0.15	94.74±0.17	92.64±0.23	88.94±0.30	**99.12±0.00**
C16	81.13±1.54	**99.75±0.50**	87.86±0.45	94.92±0.02	79.98±1.20	92.69±0.81	90.02±0.04	97.41±0.45	90.72±1.84
OA (%)	96.52±0.10	96.91±0.04	94.21±0.12	93.98±0.06	95.79±0.03	96.65±0.04	94.58±0.06	94.36±0.05	**97.44±0.11**
AA (%)	91.64±0.28	83.65±2.50	92.73±0.36	92.06±0.24	93.87±0.13	96.43±0.08	95.82±0.02	95.63±0.04	**97.26±0.35**
Kappa×100	96.04±0.12	96.48±0.05	93.40±0.14	93.12±0.07	95.21±0.04	96.18±0.04	93.82±0.07	93.89±0.05	**97.09±0.12**

performance, with an OA of 92.49%, an AA of 93.84%, and a Kappa × 100 of 91.88%, outperforming all competing methods. Although MASSFormer utilizes the integration of supplementary information to increase the richness of feature representation in the model, the proposed network still achieves gains of 2.4%, 2.39%, and 2.6%, respectively, compared to it, demonstrating its superiority in spectral–spatial feature representation and generalization. At the class level, MS^{2}CAFNet achieves 100% accuracy in multiple categories (e.g., C3, C6, C13, and C14) and maintains high accuracy in challenging classes such as C1, C8, C10, and C11, indicating its strong ability to discriminate spectrally similar or spatially mixed land-cover types. Even in categories with limited or irregular samples (e.g., C7 and C9), the model exhibits stable classification, demonstrating robustness to data imbalance and spatial complexity. As illustrated in Fig. 10, MS^{2}CAFNet produces more continuous and spatially coherent classification maps, especially in urban roads, transportation areas, and built-up regions. Compared with other state-of-the-art methods, it shows higher spatial consistency and fewer misclassifications,

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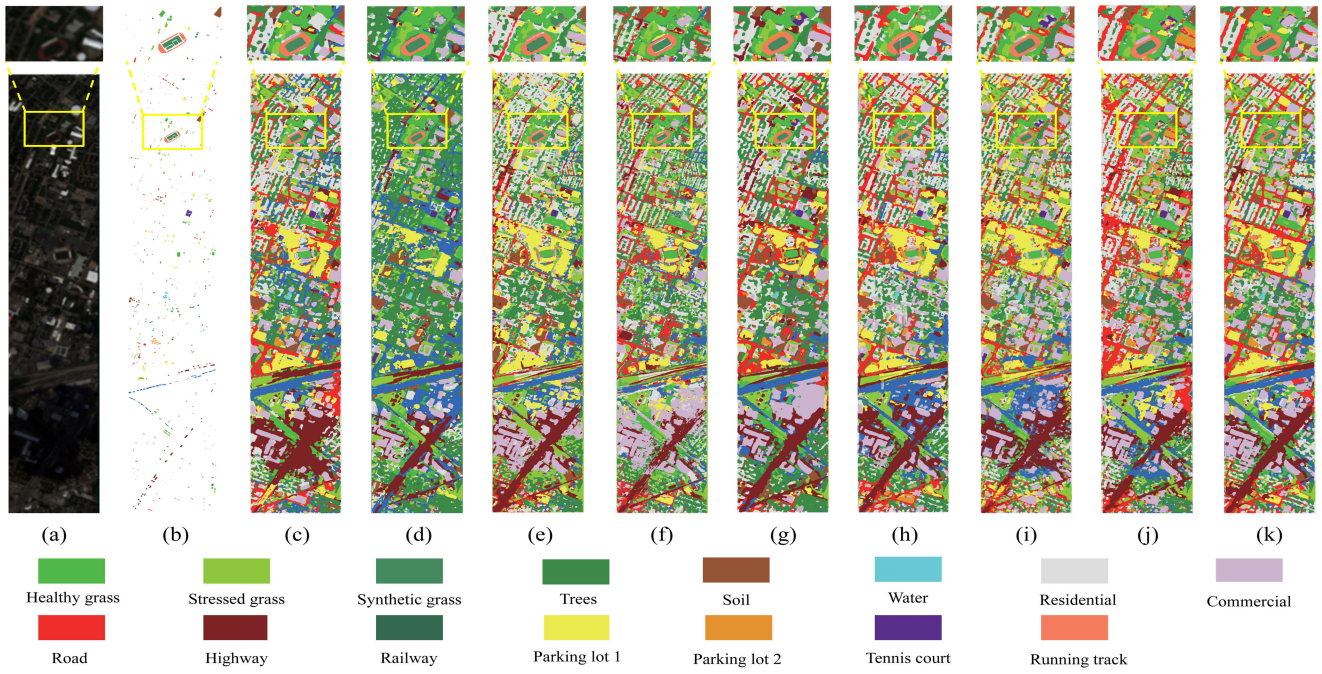


Fig. 10. Full-pixel classification maps for the Houston dataset. (a) Pseudo-color image; (b) ground-truth map; (c) MCFN; (d) DBCTNet; (e) DBMST; (f) DSNet; (g) MASSFormer; (h) CTANet; (i) DSformer; (j) MSFICNet; and (k) proposed.

TABLE VII

CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA × 100 WITH STANDARD DEVIATION FOR THE HOUSTON DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD, SECOND-BEST UNDERLINED)

NO.	MCFN	DBCTNet	DBMST	DSNet	MASSFormer	CTANet	DSformer	MSFICNet	Proposed
C1	92.72±1.30	74.24±0.19	83.81±0.15	**98.15±0.07**	91.49±0.08	81.58±0.23	92.76±0.06	84.40±0.64	93.86±0.32
C2	82.21±1.30	**100.00±0.00**	90.11±1.66	89.77±0.12	89.56±0.23	86.70±0.18	91.43±0.00	93.56±0.46	91.20±0.45
C3	**100.00±0.00**	96.65±0.05	**100.00±0.00**	99.21±0.05	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.94±0.07	**100.00±0.00**
C4	88.72±1.99	91.26±0.41	92.23±0.40	94.56±0.20	94.86±0.07	95.21±0.07	90.34±0.08	87.53±0.61	92.16±0.51
C5	98.40±0.11	89.95±0.61	98.12±0.33	98.11±0.03	92.52±0.05	97.34±0.37	91.16±0.13	92.89±0.28	**99.96±0.06**
C6	**100.00±0.00**	**100.00±0.00**	99.74±0.31	99.60±0.00	**100.00±0.00**	97.83±0.17	**100.00±0.00**	97.76±0.02	**100.00±0.00**
C7	84.14±2.65	84.03±0.63	92.11±0.73	72.69±0.19	84.55±0.12	**93.81±0.69**	81.46±0.12	86.46±0.86	88.94±0.55
C8	97.23±1.30	**100.00±0.00**	84.53±0.89	86.70±0.26	91.69±0.07	89.31±0.09	68.85±0.15	89.29±0.65	93.92±0.27
C9	78.40±0.30	73.28±1.87	92.31±2.13	70.67±0.84	82.35±0.12	64.62±0.66	78.28±0.35	91.17±0.40	86.35±0.76
C10	76.54±2.11	71.74±1.56	83.80±0.61	78.38±0.57	89.41±0.04	91.26±0.14	84.55±0.28	**92.33±1.06**	87.66±1.27
C11	89.42±1.09	94.37±0.15	91.57±1.43	73.68±0.41	86.83±0.46	84.06±0.57	83.19±0.37	87.73±1.53	**94.89±1.01**
C12	68.99±0.37	72.53±1.34	70.11±1.40	73.84±0.19	**86.56±0.26**	84.06±0.57	77.49±0.19	77.70±0.30	85.79±1.63
C13	95.42±2.29	80.04±0.40	87.54±4.42	**100.00±0.00**	89.66±0.34	87.40±0.21	33.57±0.56	73.65±2.74	**100.00±0.00**
C14	88.60±0.00	95.37±0.00	98.49±0.27	**100.00±0.00**	99.71±0.09	92.51±0.00	99.45±0.09	92.38±0.09	98.56±1.39
C15	98.93±0.25	96.41±0.07	93.97±1.66	**99.62±0.07**	92.49±0.10	93.62±0.00	88.88±0.07	99.14±0.07	94.37±0.23
OA (%)	86.94±0.14	85.39±0.46	88.37±0.29	84.88±0.12	90.09±0.06	87.13±0.09	85.37±0.01	88.87±0.08	**92.49±0.20**
AA (%)	89.31±0.12	87.99±2.98	90.56±0.45	89.00±0.07	91.45±0.03	88.96±0.06	84.09±0.04	89.73±0.12	**93.84±0.12**
Kappa×100	85.88±0.16	84.18±0.50	87.41±0.31	83.61±0.13	89.28±0.06	86.09±0.10	84.17±0.02	87.97±0.09	**91.88±0.21**

closely matching the ground-truth map. These results confirm that the proposed model effectively enhances spatial perception and discrimination in complex urban hyperspectral scenarios.

4) *Experimental Results Analysis on the Honghu Dataset:* As shown in Table VIII, the classification results of the proposed MS^{2}CAFNet and eight comparative methods are presented. The Honghu dataset contains 16 land-cover categories representing complex natural wetland scenes characterized by high interclass spectral similarity, spatial heterogeneity, and rich category diversity. The proposed method achieves the best performance across all evaluation metrics, with an OA of 98.02%, an AA of 97.57%, and a Kappa × 100 of 97.68%,

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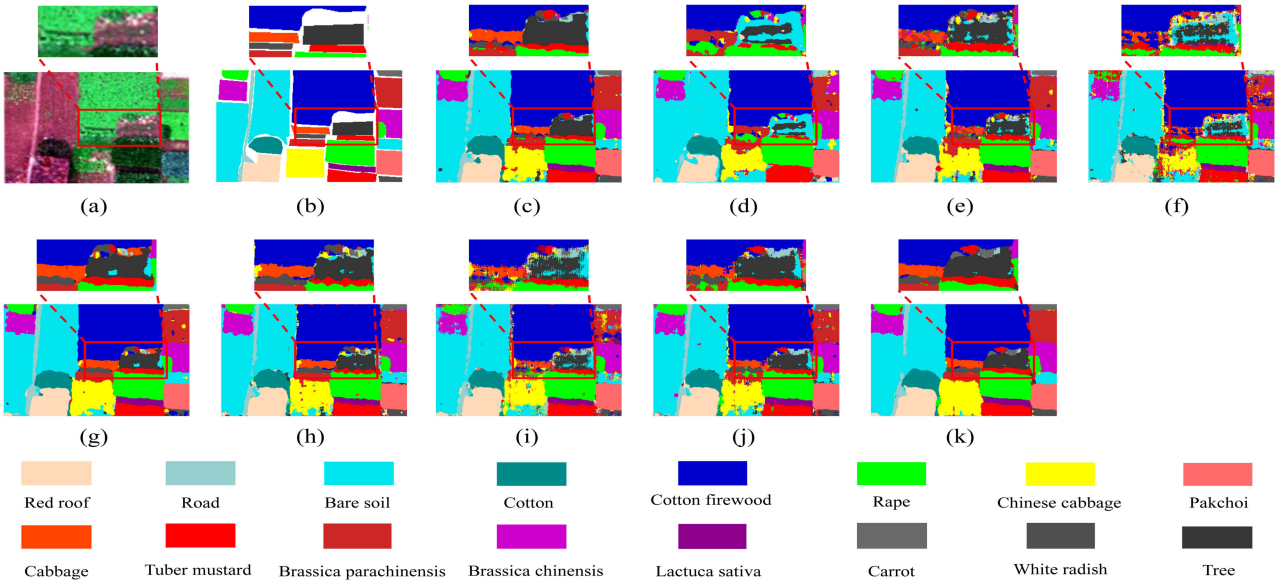


Fig. 11. Full-pixel classification maps for the Honghu dataset. (a) Pseudo-color image; (b) ground-truth map; (c) MCFN; (d) DBCTNet; (e) DBMST; (f) DSNet; (g) MASSFormer; (h) CTANet; (i) DSformer; (j) MSFICNet; and (k) proposed.

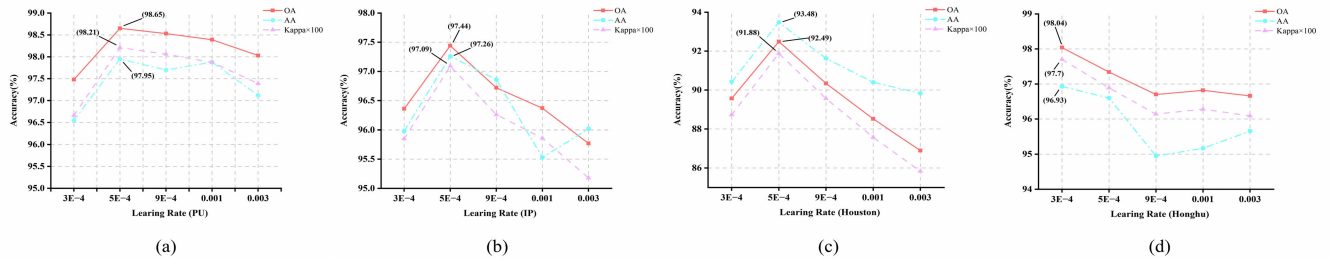
TABLE VIII

CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA × 100 WITH STANDARD DEVIATION FOR THE HONGHU DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD, SECOND-BEST UNDERLINED)

NO.	MCFN	DBCTNet	DBMST	DSNet	MASSFormer	CTANet	DSformer	MSFICNet	Proposed
C1	98.44±0.04	93.29±0.39	99.01±0.11	89.46±0.07	94.17±0.02	98.03±0.04	96.42±0.06	**99.43±0.03**	96.16±0.04
C2	71.73±1.40	46.32±3.27	87.73±0.33	65.29±0.28	87.55±0.21	**93.15±0.19**	91.54±0.22	86.56±0.29	91.94±0.12
C3	96.53±0.14	87.73±0.81	95.74±0.28	91.20±0.04	**97.98±0.04**	97.45±0.04	92.53±0.02	96.26±0.19	97.66±0.08
C4	96.95±0.32	**99.52±0.08**	98.00±0.12	99.02±0.04	98.36±0.11	98.72±0.02	98.34±0.02	98.00±0.05	98.16±0.09
C5	98.74±0.10	95.02±0.16	98.62±0.19	95.60±0.04	**99.95±0.00**	99.73±0.03	93.39±0.05	97.08±0.09	99.65±0.00
C6	95.26±0.24	92.13±0.35	92.81±0.17	89.88±0.03	**97.13±0.06**	95.18±0.28	90.96±0.03	89.29±0.17	96.70±0.08
C7	84.70±1.02	92.72±0.68	91.56±0.47	77.09±0.05	89.71±0.28	90.58±0.32	72.59±0.22	88.95±0.07	**98.99±0.04**
C8	99.41±0.07	99.76±0.07	99.21±0.23	**99.95±0.00**	98.85±0.00	98.67±0.09	**99.52±0.00**	99.63±0.02	99.13±0.01
C9	97.31±0.49	0.00±0.00	97.30±0.10	70.06±0.15	**99.07±0.11**	98.66±0.09	95.41±0.42	95.26±0.07	93.52±0.22
C10	94.73±0.13	73.93±0.17	85.15±0.34	80.53±0.15	95.78±0.10	95.63±0.14	94.09±0.03	94.16±0.30	**96.59±0.14**
C11	92.75±0.31	77.68±0.96	91.10±0.07	64.15±0.18	93.75±0.24	91.50±0.24	76.20±0.22	85.73±0.35	**96.96±0.19**
C12	97.08±0.41	96.12±0.07	97.44±0.04	91.50±0.07	**97.57±0.01**	96.92±0.06	92.23±0.04	94.28±0.15	99.65±0.04
C13	95.59±0.44	**99.66±0.09**	98.19±0.20	96.79±0.09	97.15±0.48	96.91±0.21	95.37±0.13	94.67±0.11	98.40±0.00
C14	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	94.21±0.13	98.06±0.08	**100.00±0.00**	86.97±0.36	97.96±0.44	**100.00±0.00**
C15	**100.00±0.00**	0.00±0.00	85.79±0.53	77.16±0.07	92.02±0.30	95.41±0.30	93.92±0.10	98.54±0.01	99.23±0.05
C16	91.13±0.37	94.66±1.34	91.73±0.17	80.57±0.01	96.95±0.08	85.09±0.48	88.58±0.15	95.69±0.16	**98.40±0.32**
OA (%)	95.43±0.17	90.59±0.38	95.33±0.08	88.64±0.04	96.99±0.01	96.50±0.05	91.01±0.00	94.70±0.05	**98.02±0.03**
AA (%)	94.40±0.26	78.03±2.15	94.34±0.04	85.15±0.02	95.88±0.03	95.73±0.06	91.14±0.05	94.47±0.04	**97.57±0.05**
Kappa×100	94.63±0.20	88.83±0.46	94.51±0.09	86.62±0.05	96.48±0.01	95.91±0.06	89.41±0.01	93.77±0.06	**97.68±0.04**

clearly outperforming all comparison models. In comparison to MSFICNet, which integrates the learning of local–global features and frequency features within spectral bands, our model yields respective gains of 3.32%, 3.10%, and 3.91% across the three evaluation metrics. This rigorously validates the effectiveness of the proposed mechanism in enhancing joint spectral–spatial feature modeling and discriminative capability. At the class level, MS^{2}CAFNet attains 100.00% classification accuracy in C1 and C15 and maintains high accuracy in spectrally similar or background-contaminated categories such as C2, C6, C7, C9, and C10, indicating its strong ability to separate spectrally close but semantically distinct land-cover types. For classes with limited or unevenly distributed samples (e.g., C7 and C12), the model still achieves high accuracies of 98.99% and

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(a) (b) (c) (d)

Fig. 12. Impact of different learning rates in terms of OA (%), AA (%), and Kappa $\times$ 100(%) on four datasets. (a) PU dataset. (b) IP dataset. (c) Houston dataset. (d) Honghu dataset.

99.65%, showing excellent generalization and robustness under data imbalance. As illustrated in Fig. 11, the classification map generated by MS^{2}CAFNet exhibits smoother regions and sharper, more continuous boundaries, with land-cover distributions closely matching the ground-truth map. Compared with the results of other methods, the proposed model significantly reduces misclassification and noise, producing more coherent and visually consistent results.

D. Hyperparameter Setting and Analysis

Hyperparameters are regarded as critical factors influencing the classification performance of the model, as they determine its ability to learn representative features from complex data patterns. To comprehensively evaluate the impact of hyperparameters on model performance, this section adopts a controlled-variable strategy to systematically analyze key parameters, including the learning rate, input patch size, predefined group number (g), and the number of heads (H) in the multihead attention mechanism.

1) *Learning Rate:* The learning rate is one of the most critical hyperparameters influencing the performance of deep neural networks. A larger learning rate can accelerate convergence but may cause divergence or oscillation, while a smaller learning rate ensures training stability at the cost of slower optimization. Therefore, selecting an appropriate learning rate or adopting a dynamic learning rate strategy is essential for achieving optimal classification performance and convergence stability. For the proposed method, we conducted experiments on four different datasets to investigate the effect of the learning rate on the training architecture. The learning rates are selected from a candidate set of {3E-4, 5E-4, 9E-4, 1E-3, 3E-3} to evaluate their impact on network performance. As illustrated in Fig. 12, the OA, AA, and Kappa $\times$ 100 exhibit a general trend of first increasing and then decreasing with the growth of the learning rate across most datasets. Specifically, when the learning rate is set to 5E-4, the model achieves the best performance on the PU, IP, and Houston datasets, while the highest classification accuracy on the Honghu dataset is obtained with a learning rate of 3E-4.
2) *Patch Size:* The patch size is a crucial hyperparameter that significantly affects model performance, training efficiency, and generalization ability. Enlarging the patch size strengthens the capacity of the model to acquire spatial contextual information. However, excessively large patches may introduce irrelevant background details, leading to feature confusion near class boundaries. Therefore, selecting an appropriate patch size is essential for achieving optimal classification accuracy. To evaluate the impact of patch size on the classification performance of the proposed network, experiments are conducted using candidate patch sizes of 5×5, 7×7, 9×9, 11×11, and 13×13. Fig. 13 presents the OA, AA, and Kappa $\times$ 100 obtained under different patch sizes across the four datasets. As shown in Fig. 13(a)–(c), when the patch size is set to 9×9, the proposed MS^{2}CAFNet achieves the best performance on the PU, IP, and Houston datasets. Conversely, as illustrated in Fig. 13(d), the optimal classification performance on the Honghu dataset is achieved with a patch size of 11×11. Under the same network configuration, the optimal patch size varied among datasets, which may be attributed to differences in the spatial distribution characteristics of land cover types in each scene.
3) Group number g: To evaluate the impact of the group number g in CMA on the information richness, redundancy suppression, and classification performance of cross-scale fusion, we conduct a systematic hyperparameter analysis on four datasets. The network structure and training hyperparameters are kept fixed, while only g is varied within the range of $g \in \{1, 2, 4, 8, 16\}$, and each setting is repeated five times in experiments. Fig. 14 shows the variation results of the overall OA, AA, and Kappa $\times$ 100 obtained by our method on the four datasets for different group numbers g. In general, a relatively small g tends to result in insufficient separation of cross-channel redundancy. A moderate g achieves a balance between performance and efficiency, showing the optimal performance. An excessively large g, although it improves boundary details, causes over-segmentation in several scenarios, which leads to increased fluctuations in overall performance. To ensure model generalization, the value of g is finally set to 8 in our experiments.
4) *Number of Heads H:* To evaluate the impact of the number of heads H on the feature representation capability of the MHCASA module, experimental studies are conducted on four public HSI datasets. Other structural and training hyperparameters are fixed, while H is set to vary within the range of 1,2,4,8,16. Each experimental

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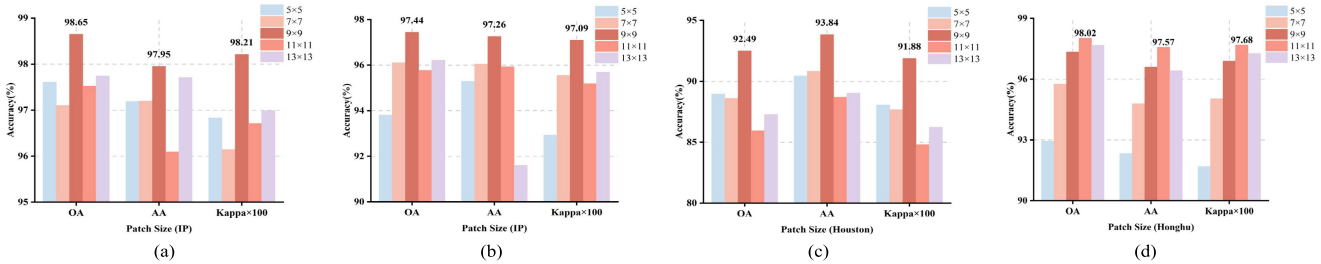


Fig. 13. Impact of different patch sizes in terms of OA (%), AA (%), and Kappa × 100(%) on four datasets. (a) PU dataset. (b) IP dataset. (c) Houston dataset. (d) Honghu dataset.

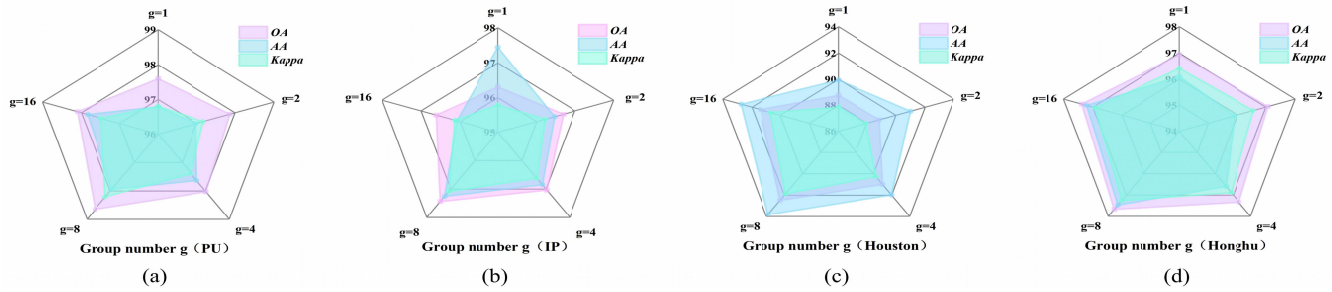


Fig. 14. Impact of different group numbers g in terms of OA (%), AA (%), and Kappa × 100(%) on four datasets. (a) PU dataset. (b) IP dataset. (c) Houston dataset. (d) Honghu dataset.

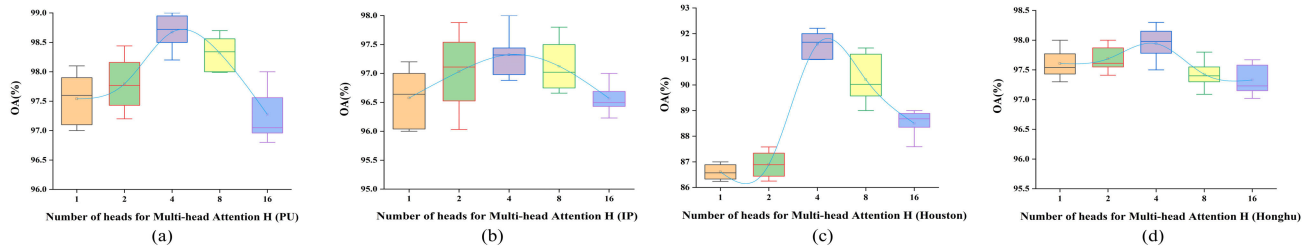


Fig. 15. Impact of different numbers of heads H in terms of OA (%) on four datasets. (a) PU dataset. (b) IP dataset. (c) Houston dataset. (d) Honghu dataset.

setting is repeated five times to reduce randomness, and the mean and standard deviation are calculated. For consistency in implementation, when the number of channels C is not divisible by H, we generated an H×d representation in the projection layer and reshaped it into a multihead structure (to ensure the dimension of each head is as balanced as possible). Fig. 15 presents the variation results of OA on the four datasets, respectively. As H increased from 1 to 4, OA showed a significant improvement, indicating that multiview global modeling is beneficial for capturing complex spatial contextual information. When H is further increased, the performance improvement declines. Finally, the value of H is set to 4 in our experiments.

E. Ablation Analysis

To systematically evaluate the contribution of each module in MS^{2}AFNet to land cover classification performance and the synergistic effect among modules, this study designed ablation experiments for different modules and validated them on four hyperspectral benchmark datasets: PU, IP, Houston, and Honghu. Specifically, the experiments constructed a series of subnetworks from Model 0 to Model 9, introducing key modules from basic feature extraction to complete fusion strategies sequentially into the network architecture. The impact mechanism of each component on the final classification accuracy was quantitatively analyzed to clarify the functional value of the core modules.

The ablation results of different model configurations are presented in Table IX. Model0 serves as the basic feature extraction network, and its baseline performance on the PU, IP, Houston, and Honghu datasets reflects the classification capability of the fundamental dual-branch multiscale feature extraction architecture without any enhancement mechanisms. Model1 introduces the channel grouping and reconstruction strategy, which slightly improves the performance, indicating that channel reorganization can alleviate feature redundancy during multiscale feature aggregation.

To analyze the effectiveness of spectral modeling, the SDGA module is gradually introduced. Model2 incorporates the similarity branch, which improves the OA to 97.36%, 95.79%, 88.23%, and 96.44%, demonstrating that similarity-driven spectral modeling helps capture global

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TABLE IX
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODULE ON THREE DATASETS (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

Cases	Component						PU	IP	Houston	Honghu
	Base	CMA	SDGA-S	SDGA-D	MHCASA	BCGF	OA (%)	OA (%)	OA (%)	OA (%)
Model0	✓						95.64	94.14	86.14	94.47
Model1	✓	✓					96.61	95.26	87.38	95.89
Model2	✓	✓	✓				97.36	95.79	88.23	96.44
Model3	✓	✓		✓			97.52	95.97	89.01	96.69
Model4	✓		✓	✓			97.85	96.32	90.15	97.03
Model5	✓	✓	✓	✓			97.98	96.73	90.36	97.34
Model6	✓				✓		98.07	96.34	90.77	97.29
Model7	✓					✓	97.04	96.13	88.45	96.85
Model8	✓	✓	✓	✓	✓	✓	98.24	96.95	91.28	97.65
Model9	✓	✓	✓	✓	✓		**98.65**	**97.44**	**92.49**	**98.02**

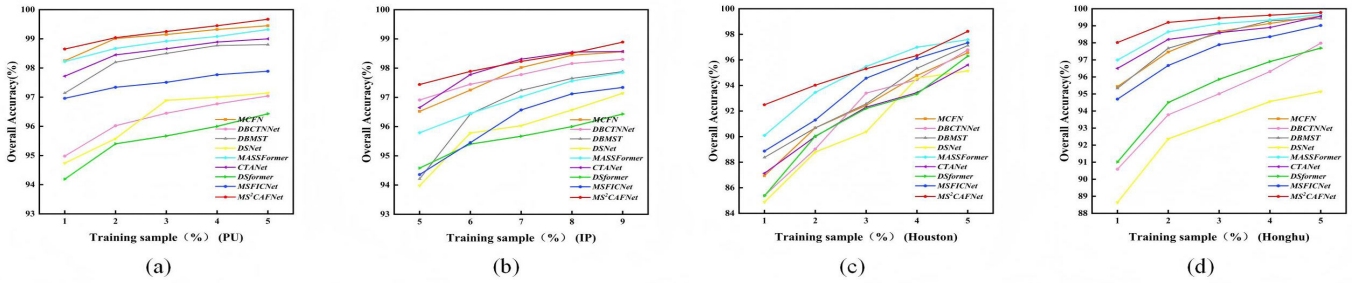


Fig. 16. Classification results of different methods on four datasets with different training samples: (a) PU dataset; (b) IP dataset; (c) Houston dataset; and (d) Honghu dataset.

spectral correlations. Model3 introduces the difference branch, further improving the OA to 97.52%, 95.97%, 89.01%, and 96.69%, indicating that difference-based modeling can enhance discriminative spectral variations. When both mechanisms are integrated (Model4 and Model5), the performance is further improved, verifying the complementary effect of similarity and difference modeling in enhancing spectral discriminability. The contribution of spatial modeling is evaluated in Model6 by introducing the MHCASA module, which raises the OA to 98.07%, 96.34%, 90.77%, and 97.29%, demonstrating the effectiveness of spatial attention in capturing spatial dependencies.

To evaluate the effect of feature fusion, Model 7 replaced the simple fusion strategy in Model1 with the proposed BCGF module. The performance improvements demonstrate that even without additional attention mechanisms, the cross-guided fusion strategy effectively enhances the fusion of spectral and spatial information. Model8 constructs a dual-branch architecture without the BCGF module, achieving 98.24%, 96.95%, 91.28%, and 97.65%, which indicates that the designed spectral and spatial branches provide strong feature representation. Finally, Model9 integrates all modules to form the complete MS^{2}CAFNet, achieving the best performance on all datasets with OA values of 98.65%, 97.44%, 92.49%, and 98.02%. These results demonstrate that the proposed spectral modeling, spatial attention, and cross-guided fusion modules provide complementary benefits and jointly improve hyperspectral classification performance.

F. Impact of Different Training Samples on The Classification Result

Due to the high cost and time-consuming nature of pixel-level annotation in HSI land cover classification, the availability of labeled samples is inherently limited. To evaluate the effectiveness of the proposed model under limited training data conditions, we conduct extensive experiments on four benchmark datasets and compare its classification performance with eight state-of-the-art methods under various training sample ratios. Specifically, 1% to 5% of labeled samples are randomly sampled for training on the PU, Houston, and Honghu datasets, while 5% to 9% are used for the IP dataset. OA is adopted as the primary evaluation metric to quantitatively assess the classification performance of the proposed method in comparison with competing approaches. The results indicate that as the number of training samples increases, the classification accuracy of all methods improves accordingly. As illustrated in Fig. 16, the proposed method consistently achieves the highest OA across all four datasets, demonstrating its strong generalization capability under limited training sample conditions.

G. Model Complexity Analysis

In order to objectively evaluate the computational efficiency of the proposed MS^{2}CAFNet and the other eight comparative algorithms, a series of experiments is conducted. In the complexity analysis, the number of parameters (Params) and

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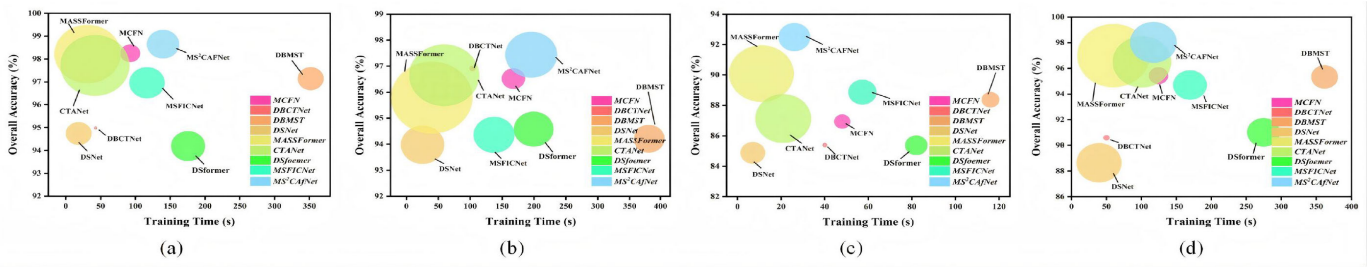


Fig. 17. Comparison of OA, parameter quantities, and average training time across four datasets. Larger bubble sizes reflect greater parameter quantities. (a) PU dataset; (b) IP dataset; (c) Houston dataset; and (d) Honghu dataset.

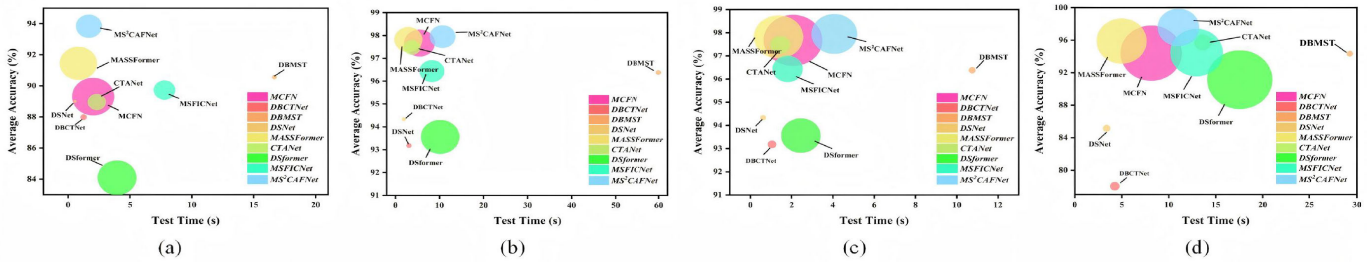


Fig. 18. Comparison of AA, FLOPs, and average test time across four datasets. Larger bubble sizes reflect greater FLOPs. (a) PU dataset; (b) IP dataset; (c) Houston dataset; and (d) Honghu dataset.



Fig. 19. Feature visualization via t-SNE on the PU dataset. (a) Input data of the PU dataset. (b) Output data of the proposed MS^{2}CAFNet on the PU dataset. (c) Output data of the MCFN on the PU dataset. (d) Output data of the DSformer on the PU dataset.

floating-point operations (FLOPs) are adopted as evaluation metrics to comprehensively assess the performance of each model. During the experiments, the batch size is set to one for all datasets. Fig. 17 illustrates the relationship among model parameters, OA, and training time on the PU, IP, Houston, and Honghu datasets. It can be observed that MS^{2}CAFNet achieves higher classification accuracy with a moderate number of parameters compared to most competing methods. Although some lightweight models have fewer parameters and shorter training time, their classification accuracy is significantly lower. In contrast, several complex models introduce a substantially larger number of parameters while providing only marginal performance gains. These results demonstrate that the proposed architecture effectively enhances feature representation without excessively increasing model complexity. Fig. 18 presents the tradeoff among FLOPs, AA, and testing time. MS^{2}CAFNet maintains competitive inference efficiency while achieving superior classification accuracy across all four datasets. Compared with more computationally expensive models, the proposed method achieves comparable or better accuracy with fewer FLOPs. Meanwhile, some methods with lower FLOPs achieve faster inference but at the expense of reduced classification performance. Overall, the proposed MS^{2}CAFNet achieves a favorable balance between computational complexity and classification accuracy. This indicates that the designed spectral attention, spatial attention, and cross-guided fusion mechanisms effectively enhance discriminative feature learning while maintaining reasonable computational overhead.

H. T-Distributed Stochastic Neighbor Embedding (t-SNE) Visual Evaluation and Analysis

To provide a more intuitive evaluation of the feature representation capability of the proposed MS^{2}CAFNet for hyperspectral imagery, the t-SNE algorithm is employed to project high-dimensional features from the PU, IP, Houston, and Honghu datasets into a 2-D space. The visualization results are shown in Figs. 19–22. In each set of images, (a) depicts the t-SNE distribution of the original input data, (b) illustrates the feature distribution generated by the proposed MS^{2}CAFNet, while (c) and (d) correspond to the feature distributions of the suboptimal and the least effective comparative models, respectively. As can be observed in (a) of each set of images, the original feature distributions of the four datasets all exhibit significant interclass overlap, with

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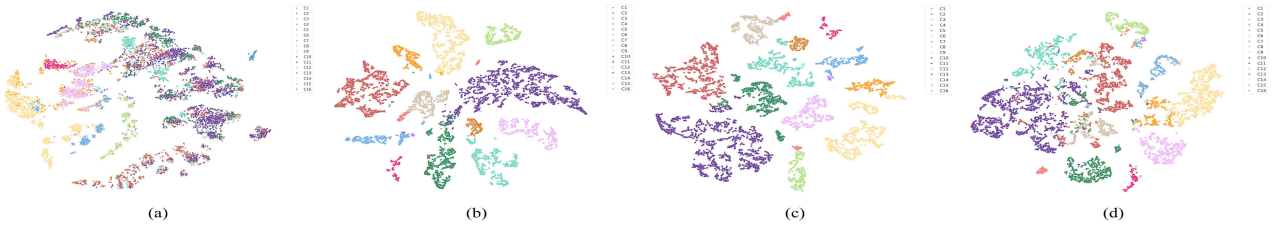


Fig. 20. Feature visualization via t-SNE on the IP dataset. (a) Input data of the IP dataset. (b) Output data of the proposed MS^{2}CAFNet on the IP dataset. (c) Output data of the DBCTNet on the IP dataset. (d) Output data of the DSNet on the IP dataset.

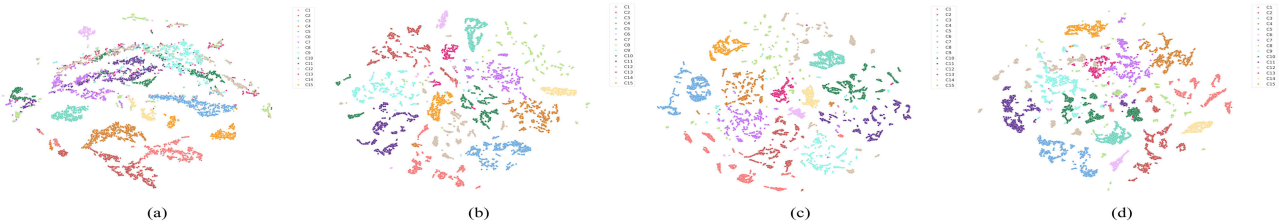


Fig. 21. Feature visualization via t-SNE on the Houston dataset. (a) Input data of the Houston dataset. (b) Output data of the proposed MS^{2}CAFNet on the Houston dataset. (c) Output data of the MASSFormer on the Houston dataset. (d) Output data of the DSNet on the Houston dataset.

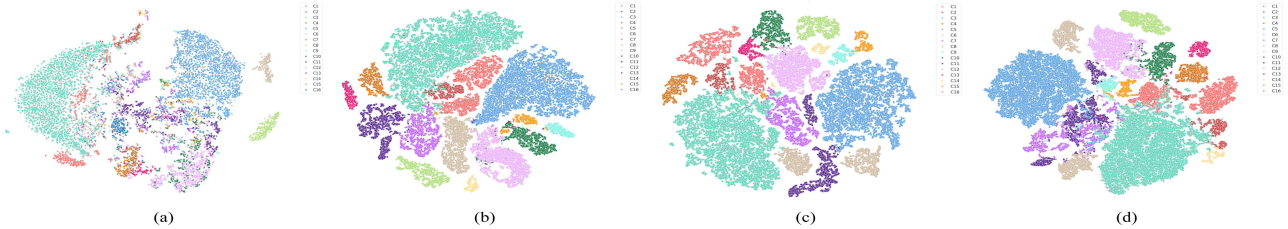


Fig. 22. Feature visualization via t-SNE on the Honghu dataset. (a) Input data of the Honghu dataset. (b) Output data of the proposed MS^{2}CAFNet on the Honghu dataset. (c) Output data of the CTANet on the Honghu dataset. (d) Output data of the DSNet on the Honghu dataset.

blurred class boundaries. As an effective feature mapping framework, the proposed MS^{2}CAFNet transforms these highly confounded features into a more discriminative representation space, yielding compact intraclass clusters and well-separated interclass margins. For example, in the PU dataset [see Fig. 19(b)], the generated feature distribution shows clear clustering patterns and distinct separability, demonstrating the strong discriminative capability of the proposed method. In contrast, DSformer [see Fig. 19(d)] still presents considerable class overlap, suggesting its limited ability in modeling spatial structures and cross-scale relationships.

V. Conclusion

In this article, an MS^{2}CAFNet is proposed for HSI classification. Different from traditional attention models, our method uses a dual-branch structure to separate spectral and spatial learning. This reduces interference between heterogeneous features. By rethinking how spectral and spatial features are extracted, organized, and interacted, MS^{2}CAFNet provides a more principled framework that aligns with the intrinsic characteristics of HSI. Specifically, the proposed DGMSA module effectively captures complementary spectral similarity and difference cues across multiple receptive fields, mitigating the over-smoothing issue commonly observed in similarity-dominant spectral attention. The MHCASA module enhances spatial modeling by jointly encoding long-range contextual dependencies, local neighborhood structures, and explicit positional information, resulting in more discriminative and spatially coherent attention responses. Moreover, the BCGF module enables alignment-driven interaction between spectral and spatial branches, ensuring semantic correspondence and coordinated enhancement across multiple scales and semantic levels. Experimental results on several widely used HSI datasets demonstrate that MS^{2}CAFNet achieves superior classification performance compared with state-of-the-art methods, particularly in challenging scenarios characterized by high interclass similarity and complex spatial heterogeneity. These results confirm the importance of coordinated spectral–spatial modeling and alignment-aware fusion strategies for advancing HSI classification. Future research will consider adapting the proposed coordination framework to lightweight network designs and transformer-based architectures, while investigating its applicability to other hyperspectral analysis tasks such as target detection and unmixing under limited training samples.

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